

# PROGETTO FINALE MODULO 2: “Sistemi operativi e linguaggi”

**Task 1:** Familiarizzazione con OS linux, shell e Command Prompt.

## Introduzione

In questo progetto si potranno osservare i procedimenti di installazione e uso di **GameShell**.

**GameShell** è un gioco eseguibile direttamente dal terminale di sistemi come Kali Linux. Il suo obiettivo è facilitare l'apprendimento e la familiarizzazione con i comandi base della Shell (come la navigazione nel filesystem e le operazioni sui file) attraverso il completamento di missioni interattive.

Durante il gioco, l'utente deve muoversi all'interno di un filesystem simulato, utilizzando i comandi della Shell, per raggiungere gli obiettivi specifici richiesti da ogni missione.

La padronanza di questi comandi è indispensabile per la navigazione, la manipolazione dei file e il raggiungimento degli obiettivi di gioco.

Di seguito è presentata una tabella che riassume i comandi essenziali che l'utente deve utilizzare e la loro funzione principale:

- **gsh goal** e **gsh check** → per visualizzare l'obiettivo e il relativo completamento (sono comandi del gioco e non della Shell)
- **cd** ( **cd ..** , **cd -** ) → per muoversi nelle diverse cartelle e sottocartelle
- **pwd** → per verificare la posizione corrente nel filesystem
- **ls** ( **ls -A** , **ls -l** ) → per verificare il contenuto di un file o directory, se nascosto (nel caso di **ls -A**) o gli eventuali permessi relativi ad un file (nel caso di **ls -l**)
- **chmod** → per cambiare i permessi di un file o directory
- **mkdir** → per la creazione di una directory
- **rm** → per la rimozione di un file o directory
- **cp** → per la copia di un file o directory
- **mv** → per spostare o rinominare un file o directory
- **alias** → per la creazione di “scorciatoie”
- **cat**, **head**, **tail** → per visualizzare un contenuto di un file
- **ps**, **pstree**, **kill**, **kill -9** → per la visualizzazione, gestione e chiusura dei processi
- **grep**, **find**, **xargs** → per la ricerca dei file
- **>** , **<** , **|** → per l'input e l'output

## Svolgimento

Per l'installazione del gioco è stato necessario l'utilizzo dei comandi `sudo apt install` e `wget` dal terminale della macchina Kali

```
(kali@kali)-[~]
$ sudo apt install gettext man-db procs psmisc nano tree bsdmainutils x11-apps wget

man-db is already the newest version (2.13.1-1).
psmisc is already the newest version (23.7-2).
psmisc set to manually installed.
x11-apps is already the newest version (7.7+11+b1).
x11-apps set to manually installed.
wget is already the newest version (1.25.0-2).
Upgrading:
  gettext-base liblzma5 libproc2-0 libxml2-dev libxml2-utils nano procs tree xz-utils

Installing:
  bsdmainutils gettext

Installing dependencies:
  liblzma-dev libxml2-16 ncal

Suggested packages:
  calendar vacation mailutils autopoint gettext-doc libasprintf-dev libgettextpo-dev gnutlib-l10n liblzma-doc

Summary:
  Upgrading: 9, Installing: 5, Removing: 0, Not Upgrading: 1456
  Download size: 6,453 kB
  Space needed: 9,506 kB / 62.2 GB available

Continue? [Y/n] y
Get:1 http://kali.download/kali kali-rolling/main amd64 liblzma5 amd64 5.8.1-2 [310 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 nano amd64 8.6-1 [647 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 libproc2-0 amd64 2:4.0.4-9 [65.6 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 procs amd64 2:4.0.4-9 [882 kB]
Get:6 http://kali.download/kali kali-rolling/main amd64 xz-utils amd64 5.8.1-2 [660 kB]
Get:9 http://http.kali.org/kali kali-rolling/main amd64 libxml2-16 amd64 2.14.6+dfsg-0.1 [651 kB]
Get:10 http://http.kali.org/kali kali-rolling/main amd64 gettext amd64 0.23.1-2+b1 [1,680 kB]
Get:11 http://kali.download/kali kali-rolling/main amd64 liblzma-dev amd64 5.8.1-2 [349 kB]
Get:12 http://http.kali.org/kali kali-rolling/main amd64 libxml2-dev amd64 2.14.6+dfsg-0.1 [756 kB]
Get:13 http://http.kali.org/kali kali-rolling/main amd64 libxml2-utils amd64 2.14.6+dfsg-0.1 [123 kB]
Get:14 http://kali.download/kali kali-rolling/main amd64 tree amd64 2.2.1-4 [59.8 kB]
Get:5 http://http.kali.org/kali kali-rolling/main amd64 gettext-base amd64 0.23.1-2+b1 [244 kB]
Get:7 http://kali.mirror.garr.it/kali kali-rolling/main amd64 ncal amd64 12.1.8 [19.7 kB]
Get:8 http://kali.mirror.garr.it/kali kali-rolling/main amd64 bsdmainutils all 12.1.8 [5,952 B]
Fetched 6,453 kB in 1s (6,299 kB/s)
(Reading database ... 415658 files and directories currently installed.)
Preparing to unpack .../liblzma5_5.8.1-2_amd64.deb ...
Unpacking liblzma5:amd64 (5.8.1-2) over (5.8.1-1) ...

(kali@kali)-[~]
$ wget https://github.com/phyver/GameShell/releases/download/latest/gameshell.sh
--2025-11-13 14:16:19-- https://github.com/phyver/GameShell/releases/download/latest/gameshell.sh
Resolving github.com (github.com)... 140.82.121.4
Connecting to github.com (github.com)|140.82.121.4|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/94422621/59962d4f-cbe9-4183-a63a-8d4513f145d4?sp=r
sh0rct-application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2025-11-13T1
ERY%2FOEFkSebA%3D6jwT=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXN0dWUiY29tIiwiaXVkiIjoicmVsZWZzZS1hc3NldHMuz2L0aHVidXNlcmVubnRlbn
cHJvZHVjdGlvbi5ibG9iLmVucmUud2luZG93cy5uZXQifQ.5fsXiLafx_wY8tn3cAVBrx2xcS20XhYfm7w8dR3wBgM6response-content-disposition=attachment%3B%20f
me%3Dgameshell.sh0rct-application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&sk
PT0V5jyFMEtM6tgERY%2FOEFkSebA%3D6jwT=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXN0dWUiY29tIiwiaXVkiIjoicmVsZWZzZS1hc3NldHMuz2L0aHV
mVsZWZzZWZzZ2V0chJvZHVjdGlvbi5ibG9iLmVucmUud2luZG93cy5uZXQifQ.5fsXiLafx_wY8tn3cAVBrx2xcS20XhYfm7w8dR3wBgM6response-content-disposition=at
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.111.133, 185.199.108.133, 185.199.109.13
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 209794 (205K) [application/octet-stream]
Saving to: 'gameshell.sh'

gameshell.sh
100%[-----]

2025-11-13 14:16:20 (9.68 MB/s) - 'gameshell.sh' saved [209794/209794]
```

Per l'avvio del gioco è stato inoltre necessario un cambio di permessi così da poter rendere il file eseguibile dall'utente.

```

kali@kali: ~
File Actions Edit View Help
(kali@kali)-[~]
$ ls
a.out Desktop Documents «dos» dos Downloads gameshell.sh Music nano.15139.save Net-DNS-1.37 Net-DNS-1.37.tar.gz Net-DNS-1.37.tar.gz
(kali@kali)-[~]
$ chmod +x gameshell.sh
(kali@kali)-[~]
$
(kali@kali)-[~]
$ ls
a.out Desktop Documents «dos» dos Downloads gameshell.sh Music nano.15139.save Net-DNS-1.37 Net-DNS-1.37.tar.gz Net-DNS-1.37.tar.gz
(kali@kali)-[~]
$

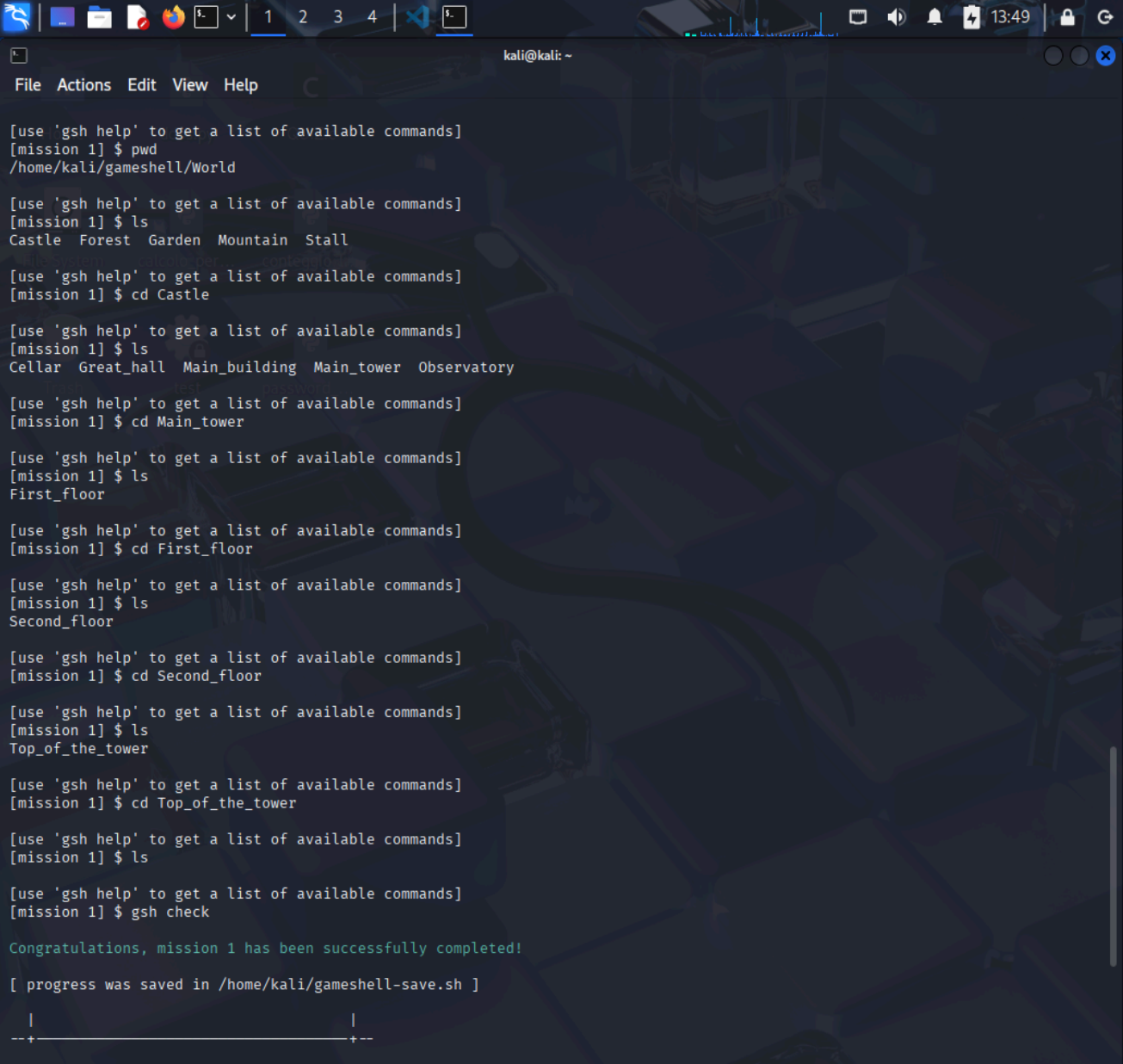
```

Successivamente si è proceduto ad avviare il gioco attraverso il comando **./gameshell.sh** e la schermata che l'utente visualizza all' avvio è la seguente:



## Missione n. 1:

Si richiede all'utente di raggiungere il tetto della torre, muovendosi all'interno del castello attraverso i comandi **cd** e **ls**.

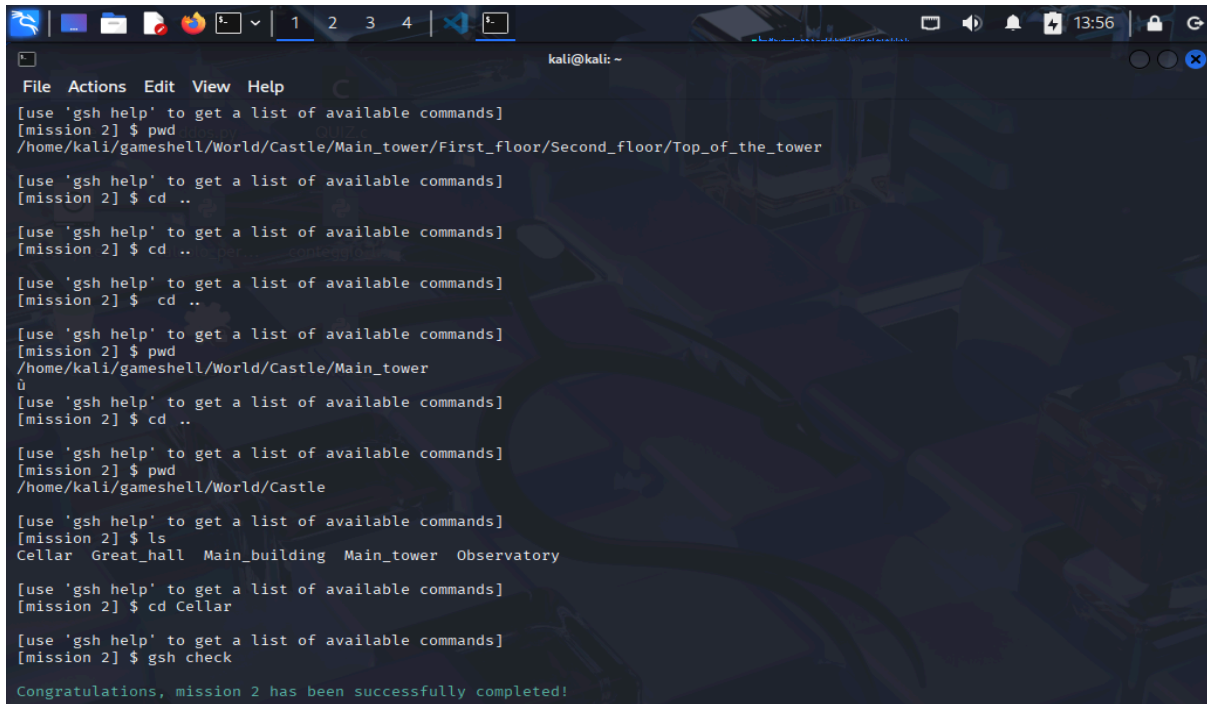


```
kali@kali: ~  
File Actions Edit View Help  
[use 'gsh help' to get a list of available commands]  
[mission 1] $ pwd  
/home/kali/gameshell/World  
[use 'gsh help' to get a list of available commands]  
[mission 1] $ ls  
Castle Forest Garden Mountain Stall  
[use 'gsh help' to get a list of available commands]  
[mission 1] $ cd Castle  
[use 'gsh help' to get a list of available commands]  
[mission 1] $ ls  
Cellar Great_hall Main_building Main_tower Observatory  
[use 'gsh help' to get a list of available commands]  
[mission 1] $ cd Main_tower  
[use 'gsh help' to get a list of available commands]  
[mission 1] $ ls  
First_floor  
[use 'gsh help' to get a list of available commands]  
[mission 1] $ cd First_floor  
[use 'gsh help' to get a list of available commands]  
[mission 1] $ ls  
Second_floor  
[use 'gsh help' to get a list of available commands]  
[mission 1] $ cd Second_floor  
[use 'gsh help' to get a list of available commands]  
[mission 1] $ ls  
Top_of_the_tower  
[use 'gsh help' to get a list of available commands]  
[mission 1] $ cd Top_of_the_tower  
[use 'gsh help' to get a list of available commands]  
[mission 1] $ ls  
[use 'gsh help' to get a list of available commands]  
[mission 1] $ gsh check  
Congratulations, mission 1 has been successfully completed!  
[ progress was saved in /home/kali/gameshell-save.sh ]  
|  
--+-----+--
```



## Missione n. 2:

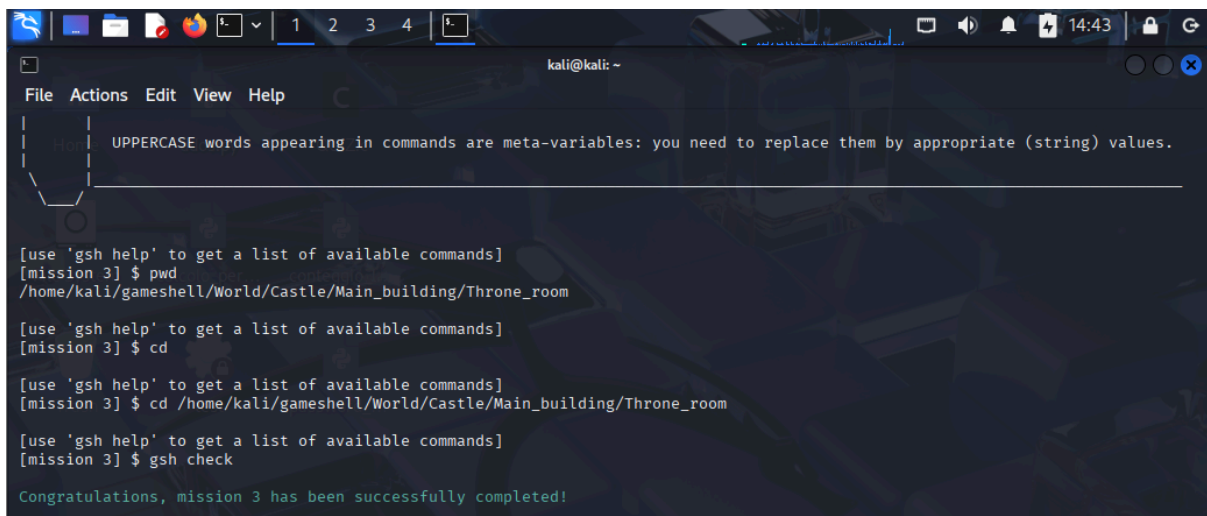
Si richiede all'utente di tornare indietro e di muoversi fino alla cartella "Cellar" tramite i comandi **pwd** (per vedere la posizione corrente) e **cd ..** (per tornare indietro e risalire nella cartella precedente).



```
kali@kali: ~  
File Actions Edit View Help  
[use 'gsh help' to get a list of available commands]  
[mission 2] $ pwd  
/home/kali/gameshell/World/Castle/Main_tower/First_floor/Second_floor/Top_of_the_tower  
  
[use 'gsh help' to get a list of available commands]  
[mission 2] $ cd ..  
  
[use 'gsh help' to get a list of available commands]  
[mission 2] $ cd ..  
  
[use 'gsh help' to get a list of available commands]  
[mission 2] $ cd ..  
  
[use 'gsh help' to get a list of available commands]  
[mission 2] $ pwd  
/home/kali/gameshell/World/Castle/Main_tower  
ù  
[use 'gsh help' to get a list of available commands]  
[mission 2] $ cd ..  
  
[use 'gsh help' to get a list of available commands]  
[mission 2] $ pwd  
/home/kali/gameshell/World/Castle  
  
[use 'gsh help' to get a list of available commands]  
[mission 2] $ ls  
Cellar Great_hall Main_building Main_tower Observatory  
  
[use 'gsh help' to get a list of available commands]  
[mission 2] $ cd Cellar  
  
[use 'gsh help' to get a list of available commands]  
[mission 2] $ gsh check  
  
Congratulations, mission 2 has been successfully completed!
```

## Missione n. 3:

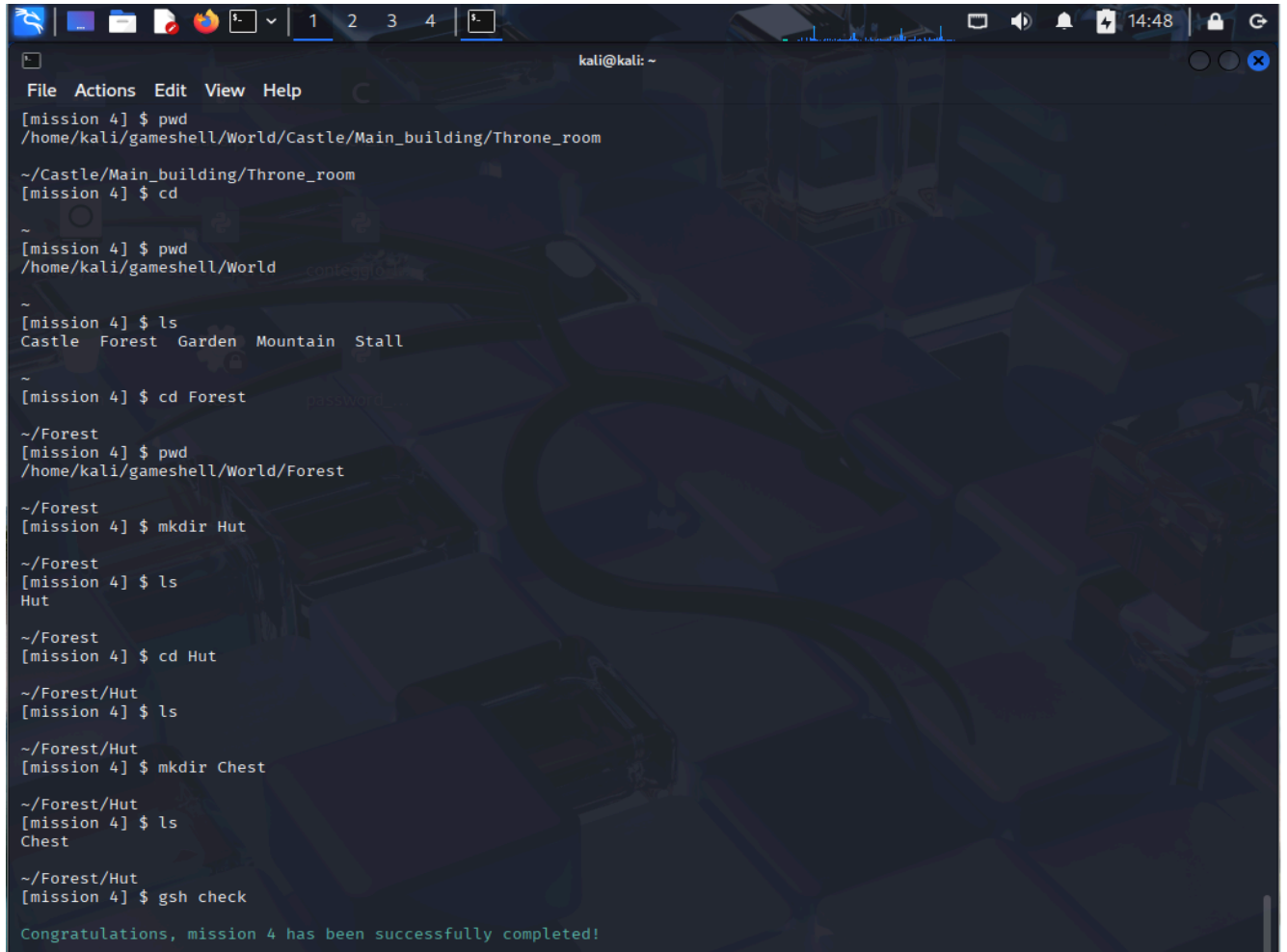
Si richiede all'utente di tornare al punto di partenza tramite il comando **cd** e di raggiungere la "Throne\_room" tramite un solo comando.



```
kali@kali: ~  
File Actions Edit View Help  
UPPERCASE words appearing in commands are meta-variables: you need to replace them by appropriate (string) values.  
  
[use 'gsh help' to get a list of available commands]  
[mission 3] $ pwd  
/home/kali/gameshell/World/Castle/Main_building/Throne_room  
  
[use 'gsh help' to get a list of available commands]  
[mission 3] $ cd  
  
[use 'gsh help' to get a list of available commands]  
[mission 3] $ cd /home/kali/gameshell/World/Castle/Main_building/Throne_room  
  
[use 'gsh help' to get a list of available commands]  
[mission 3] $ gsh check  
  
Congratulations, mission 3 has been successfully completed!
```

## Missione n. 4:

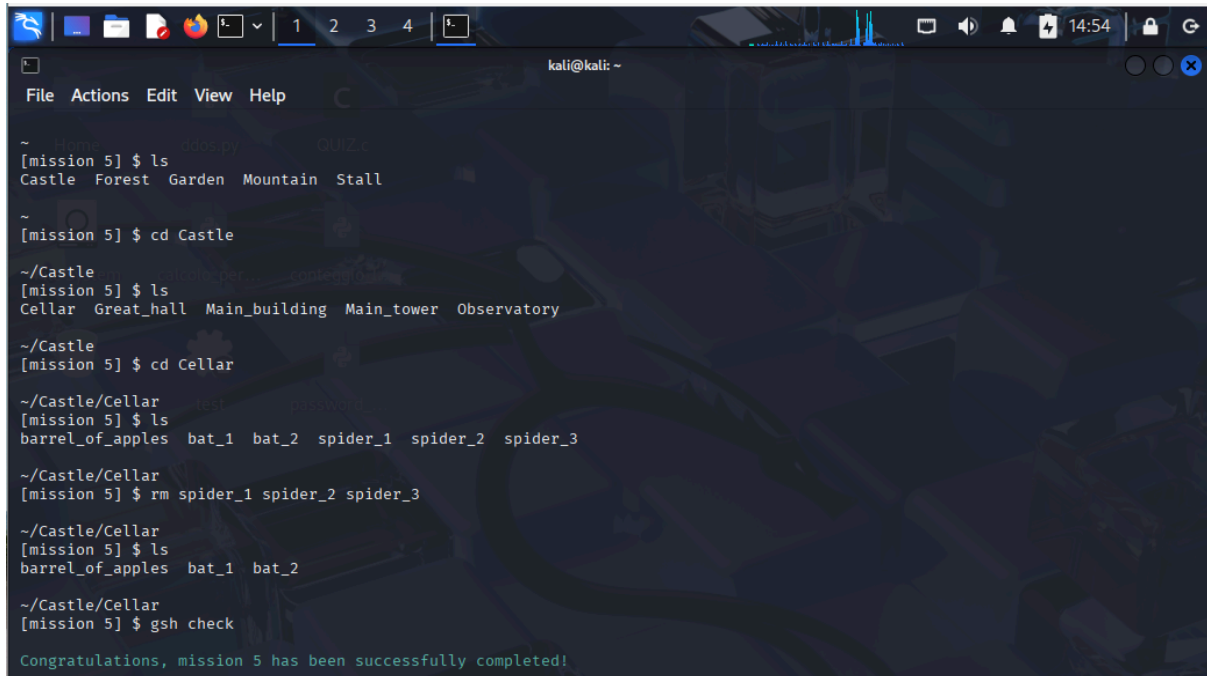
Si richiede all'utente di raggiungere la cartella "Forest", utilizzando i comandi **cd** ed **ls** (per visualizzare il contenuto delle directory) e di creare, tramite il comando **mkdir**, una sottocartella "Hut" con all'interno una sottocartella "Chest".



```
kali@kali: ~  
File Actions Edit View Help  
[mission 4] $ pwd  
/home/kali/gameshell/World/Castle/Main_building/Throne_room  
~/Castle/Main_building/Throne_room  
[mission 4] $ cd  
~  
[mission 4] $ pwd  
/home/kali/gameshell/World  
~  
[mission 4] $ ls  
Castle Forest Garden Mountain Stall  
~  
[mission 4] $ cd Forest  
~/Forest  
[mission 4] $ pwd  
/home/kali/gameshell/World/Forest  
~/Forest  
[mission 4] $ mkdir Hut  
~/Forest  
[mission 4] $ ls  
Hut  
~/Forest  
[mission 4] $ cd Hut  
~/Forest/Hut  
[mission 4] $ ls  
~/Forest/Hut  
[mission 4] $ mkdir Chest  
~/Forest/Hut  
[mission 4] $ ls  
Chest  
~/Forest/Hut  
[mission 4] $ gsh check  
Congratulations, mission 4 has been successfully completed!
```

## Missione n.5:

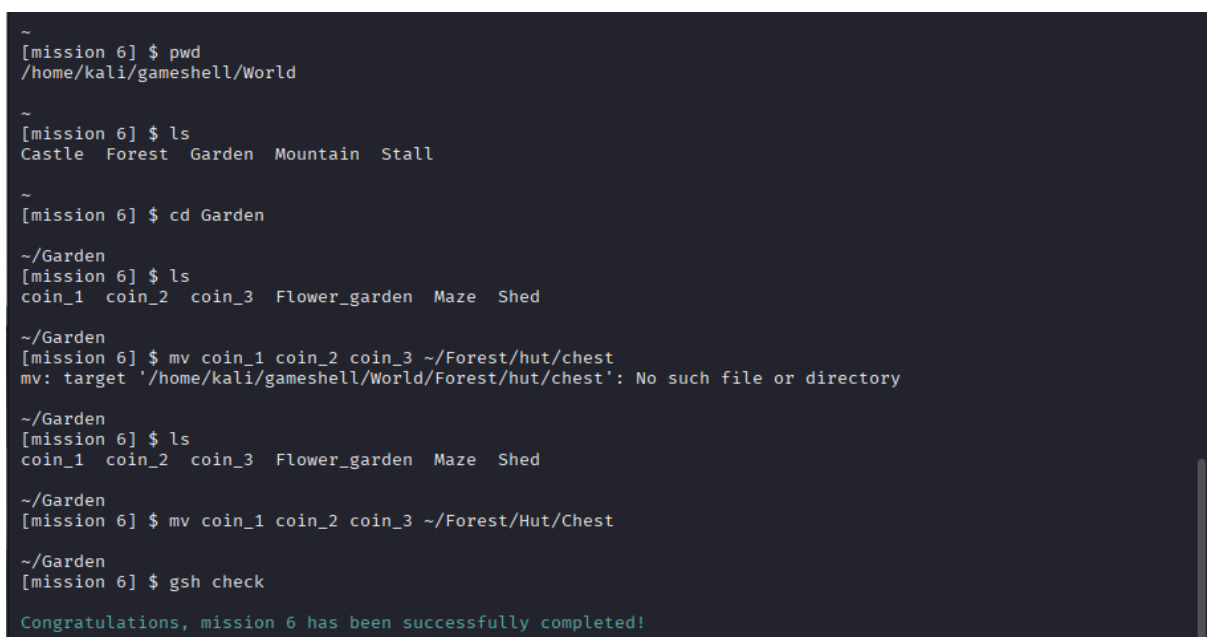
Si richiede all'utente di raggiungere la cartella "Cellar" e, tramite il comando **rm**, rimuovere tutti gli "spiders", lasciando i "bats" da soli.



```
kali@kali: ~  
File Actions Edit View Help  
~  
[mission 5] $ ls  
Castle Forest Garden Mountain Stall  
~  
[mission 5] $ cd Castle  
~/Castle  
[mission 5] $ ls  
Cellar Great_hall Main_building Main_tower Observatory  
~/Castle  
[mission 5] $ cd Cellar  
~/Castle/Cellar  
[mission 5] $ ls  
barrel_of_apples bat_1 bat_2 spider_1 spider_2 spider_3  
~/Castle/Cellar  
[mission 5] $ rm spider_1 spider_2 spider_3  
~/Castle/Cellar  
[mission 5] $ ls  
barrel_of_apples bat_1 bat_2  
~/Castle/Cellar  
[mission 5] $ gsh check  
Congratulations, mission 5 has been successfully completed!
```

## Missione n. 6:

Si richiede all'utente di spostare i "coins" all'interno della cartella "Chest" tramite il comando **mv**.



```
~  
[mission 6] $ pwd  
/home/kali/gameshell/World  
~  
[mission 6] $ ls  
Castle Forest Garden Mountain Stall  
~  
[mission 6] $ cd Garden  
~/Garden  
[mission 6] $ ls  
coin_1 coin_2 coin_3 Flower_garden Maze Shed  
~/Garden  
[mission 6] $ mv coin_1 coin_2 coin_3 ~/Forest/hut/chest  
mv: target '/home/kali/gameshell/World/Forest/hut/chest': No such file or directory  
~/Garden  
[mission 6] $ ls  
coin_1 coin_2 coin_3 Flower_garden Maze Shed  
~/Garden  
[mission 6] $ mv coin_1 coin_2 coin_3 ~/Forest/Hut/Chest  
~/Garden  
[mission 6] $ gsh check  
Congratulations, mission 6 has been successfully completed!
```

## Missione n. 7:

Si richiede all'utente di visualizzare i "coins" nascosti con il comando **ls -A** e successivamente di spostarli con il comando **mv**.

```
~
[mission 7] $ pwd
/home/kali/gameshell/World

~
[mission 7] $ cd Garden

~/Garden
[mission 7] $ ls -A
.50110_coin_2 .57346_coin_1 .59507_coin_3 Flower_garden Maze Shed

~/Garden
[mission 7] $ mv .50110_coin_2 .57346_coin_1 .59507_coin_3 ~/Forest/Hut/Chest

~/Garden
[mission 7] $ gsh check

Congratulations, mission 7 has been successfully completed!
```

## Missione n. 8:

Si richiede all'utente di raggiungere "Cellar" e di eliminare tutti gli "spider". Per eseguire questo comando è necessario l'uso del comando **rm**, seguito da asterischi (\*) prima e dopo della parola "spider", così da consentire l'eliminazione di tutti quei file che contengono quella parola. Il comando è il seguente: **rm \*spider\***.

```
kali@kali: ~
File Actions Edit View Help

~
[mission 8] $ pwd
/home/kali/gameshell/World

~
[mission 8] $ ls
Castle Forest Garden Mountain Stall

~
[mission 8] $ cd Castle

~/Castle
[mission 8] $ ls
Cellar Great_hall Main_building Main_tower Observatory

~/Castle
[mission 8] $ cd Cellar

~/Castle/Cellar
[mission 8] $ ls
11034_bat_3 14127_spider_21 18667_spider_20 22206_spider_1 23479_spider_48 26333_spider_22 292_spider_40 30910_spider_42
12049_spider_5 14668_spider_16 19880_spider_32 22249_spider_18 24614_spider_9 26657_spider_43 29468_spider_11 30946_bat_1
12663_spider_34 1622_spider_28 20605_spider_41 23008_spider_13 24740_spider_33 26746_spider_37 29517_spider_6 31391_spider_4
12800_spider_10 16756_spider_27 20625_spider_19 2321_spider_29 25556_spider_15 27210_spider_46 29687_spider_47 32038_spider_38
13162_spider_24 17615_bat_2 20930_bat_4 23371_spider_14 25937_spider_45 28022_spider_36 30530_spider_8 32308_spider_26

~/Castle/Cellar
[mission 8] $ rm *spider
rm: cannot remove '*spider': No such file or directory

~/Castle/Cellar
[mission 8] $ rm *spider*

~/Castle/Cellar
[mission 8] $ ls
11034_bat_3 17615_bat_2 20930_bat_4 30946_bat_1 4359_bat_5 barrel_of_apples

~/Castle/Cellar
[mission 8] $ gsh check

Congratulations, mission 8 has been successfully completed!
```

## Missione n. 9:

Si richiede all'utente di rimuovere, con il comando **rm**, tutti i file nascosti (visibili con il comando **ls -A**) contenenti la parola "spider", dunque, sempre attraverso l'utilizzo degli asterischi (\*) prima e dopo la parola "spider".

```
kali@kali: ~  
File Actions Edit View Help  
~  
~/Castle/Cellar  
[mission 9] $ ls -A  
11034_bat_3 .13248_spider_22 .15256_spider_9 .19046_spider_46 20930_bat_4 .22107_spider_47 .24422_spider_45 .28030_sp  
.11125_spider_39 .14088_spider_26 .15502_spider_42 .19430_bat_5 .2100_spider_35 .22123_spider_36 .25078_bat_1 .28053_sp  
.11750_spider_34 .14477_spider_43 .15799_spider_18 .1958_spider_50 .21125_spider_44 .23275_spider_29 .25267_spider_33 .28100_sp  
.12758_spider_2 .14752_bat_2 17615_bat_2 .20041_spider_20 .21748_spider_31 .23852_spider_13 .25513_spider_38 .28311_sp  
.13016_spider_30 .1480_spider_41 .18997_spider_23 .20899_spider_7 .22020_spider_16 .24324_spider_10 .27170_spider_5 .28375_sp  
~/Castle/Cellar  
[mission 9] $ rm *spider*  
rm: cannot remove '*spider*': No such file or directory  
~/Castle/Cellar  
[mission 9] $ rm .*spider*  
~/Castle/Cellar  
[mission 9] $ ls  
11034_bat_3 17615_bat_2 20930_bat_4 30946_bat_1 4359_bat_5 barrel_of_apples  
~/Castle/Cellar  
[mission 9] $ goal check  
Command 'goal' not found, did you mean:  
  command 'gcal' from deb gcal  
  command 'gdal' from deb gdal-bin  
  command 'gol' from deb growl-for-linux  
Try: sudo apt install <deb name>  
~/Castle/Cellar  
[mission 9] $ gsh check  
Congratulations, mission 9 has been successfully completed!
```

## Missione n. 10:

Si richiede all'utente di raggiungere la cartella "Great\_hall" e di copiare, con il comando **cp**, i file standard nella cartella "Chest".

```
~  
[mission 10] $ cd Castle  
~/Castle  
[mission 10] $ ls  
Cellar/ Great_hall/ Main_building/ Main_tower/ Observatory/  
~/Castle  
[mission 10] $ cd Great_hall  
~/Castle/Great_hall  
[mission 10] $ ls  
25933_suit_of_armour 39363_stag_head 3994_decorative_shield standard_1 standard_2 standard_3 standard_4  
~/Castle/Great_hall  
[mission 10] $ cp standard_1 standard_2 standard_3 standard_4 ~/Forest/Hut/Chest  
~/Castle/Great_hall  
[mission 10] $ gsh check  
Congratulations, mission 10 has been successfully completed!
```

## Missione n. 11:

Si richiede all'utente di copiare, con il comando **cp**, tutti i file contenenti la parola "tapestry" dalla cartella "Great\_hall" nella cartella "Chest". Il comando necessario è: **cp \*tapestry\*** (la parola all'interno dei 2 asterischi per copiare tutti i file contenenti quella parola).

```
~/Castle/Great_hall
[mission 11] $ ls
1276_tapestry_08      24984_tapestry_05    46072_tapestry_06    48931_tapestry_09    51160_decorative_shield  54965_tapestry_07    8435_tapestry_03
24825_suit_of_armour  36752_tapestry_02    48779_stag_head      50099_tapestry_10    54707_tapestry_04      62389_tapestry_01    standard_1

~/Castle/Great_hall
[mission 11] $ cp *tapestry* ~/Forest/Hut/Chest

~/Castle/Great_hall
[mission 11] $ gsh check

Congratulations, mission 11 has been successfully completed!
```

## Missione n. 12:

Si richiede all'utente di raggiungere la cartella "First\_floor" e di copiare, col comando **cp**, il file più vecchio presente al suo interno. Necessario qui l'utilizzo del comando **ls -l** per visualizzare le info dei file presenti.

```
~/Castle
[mission 12] $ cd Main_tower

~/Castle/Main_tower
[mission 12] $ ls
First_floor/

~/Castle/Main_tower
[mission 12] $ cd First_floor

~/Castle/Main_tower/First_floor
[mission 12] $ ls
painting_cZMfMiVe  painting_MIHsbHDY  painting_OwxnQnVz  Second_floor/

~/Castle/Main_tower/First_floor
[mission 12] $ ls -l
total 16
-rw-rw-r-- 1 kali kali 1455 Oct  7 1995 painting_cZMfMiVe
-rw-rw-r-- 1 kali kali 1055 Sep  9 2013 painting_MIHsbHDY
-rw-rw-r-- 1 kali kali 1503 Dec 16 1989 painting_OwxnQnVz
drwxrwxr-x 3 kali kali 4096 Nov 13 14:36 Second_floor/

~/Castle/Main_tower/First_floor
[mission 12] $ cp painting_OwxnQnVz ~/Forest/Hut/Chest

~/Castle/Main_tower/First_floor
[mission 12] $ gsh check

Congratulations, mission 12 has been successfully completed!
```



## Missione n. 14:

Si richiede all'utente di creare una scorciatoia tramite il comando **alias**. In particolare, attraverso il codice: **alias la='ls -A'**. In questo caso, il comando che prima era eseguibile tramite **ls -A**, viene sostituito con **la**.

```
~  
[mission 45] $ alias la='ls -A'  
  
~  
[mission 45] $ la  
.bashrc  Castle/  Forest/  Garden/  .gshrc  .lessht  .local/  Mountain/  .nice_rock  Stall/  
  
~  
[mission 45] $
```

## Missione n. 15:

Si richiede all'utente di raggiungere la cartella "Chest" e di creare al suo interno un file di testo chiamato "journal.txt" con il comando **nano**.

```
[mission 15] $ pwd  
/home/kali/gameshell/World  
  
~  
[mission 15] $ cd ~/Forest/Hut/Chest  
  
~/Forest/Hut/Chest  
[mission 15] $ pwd  
/home/kali/gameshell/World/Forest/Hut/Chest  
  
~/Forest/Hut/Chest  
[mission 15] $ ls  
1276_tapestry_08  36752_tapestry_02  48931_tapestry_09  54707_tapestry_04  62389_tapestry_01  coin_1  coin_3  standard_1  standard_3  
24984_tapestry_05  46072_tapestry_06  50099_tapestry_10  54965_tapestry_07  8435_tapestry_03  coin_2  painting_OwxnQnVz  standard_2  standard_4  
  
~/Forest/Hut/Chest  
[mission 15] $ nano journal.txt  
  
~/Forest/Hut/Chest  
[mission 15] $ gsh check  
  
Congratulations, mission 15 has been successfully completed!
```

## Missione n. 16:

Si richiede all'utente, tramite il comando **alias**, di creare la scorciatoia **journal** per il file **journal.txt**

```
~/Forest/Hut/Chest  
[mission 16] $ ls  
1276_tapestry_08  36752_tapestry_02  50099_tapestry_10  62389_tapestry_01  8435_tapestry_03  coin_3  Gold_Coin_2  00000_silver_coin_00000  standard_2  
17323  46072_tapestry_06  54707_tapestry_04  7c848efd78a  coin_1  crown  journal.txt  painting_OwxnQnVz  standard_3  
24984_tapestry_05  48931_tapestry_09  54965_tapestry_07  7fcd33052e8f71ca6247d62ada1  coin_2  gold_coin_1  00000_copper_coin_00000  standard_1  standard_4  
  
~/Forest/Hut/Chest  
[mission 16] $ alias journal='nano ~/Forest/Hut/Chest/journal.txt'  
  
~/Forest/Hut/Chest  
[mission 16] $ gsh check  
  
Congratulations, mission 16 has been successfully completed!
```

### **Missione n. 17:**

Si richiede all'utente di eliminare il file che contiene la parola "spider\_queen". In questo livello è presente un timer per il quale bisogna raggiungere l'obiettivo entro 20 secondi. Utile a velocizzare il processo di ricerca del file è il **Tab** che, se premuto dopo aver scritto la prima lettera, completa la parola.

```
~/Castle/Cellar
[mission 17] $ cd .Lair_of_the_spider_queen\ eoHomoBTDZFLCYxN FmPlJFTHCTCmYPSX/

~/Castle/Cellar/.Lair_of_the_spider_queen eoHomoBTDZFLCYxN FmPlJFTHCTCmYPSX
[mission 17] $ la
jtOxnfvkNcnnzYMX_spider_queen_DoVhWazZRzTQxxs WVVItchooEqdkyTF_baby_bat_DatMNVWTqRQCnRw

~/Castle/Cellar/.Lair_of_the_spider_queen eoHomoBTDZFLCYxN FmPlJFTHCTCmYPSX
[mission 17] $ rm jtOxnfvkNcnnzYMX_spider_queen_DoVhWazZRzTQxxs

~/Castle/Cellar/.Lair_of_the_spider_queen eoHomoBTDZFLCYxN FmPlJFTHCTCmYPSX
[mission 17] $ la
WVVItchooEqdkyTF_baby_bat_DatMNVWTqRQCnRw

~/Castle/Cellar/.Lair_of_the_spider_queen eoHomoBTDZFLCYxN FmPlJFTHCTCmYPSX
[mission 17] $ goal check
Command 'goal' not found, did you mean:
  command 'gol' from deb growl-for-linux
  command 'gdal' from deb gdal-bin
  command 'gcal' from deb gcal
Try: sudo apt install <deb name>

~/Castle/Cellar/.Lair_of_the_spider_queen eoHomoBTDZFLCYxN FmPlJFTHCTCmYPSX
[mission 17] $ gsh check
Perfect, it took you only 17 seconds to complete this mission!

Congratulations, mission 17 has been successfully completed!
```

## Missione n. 19

Si richiede all'utente di utilizzare il comando **flarigo** per eseguire dei fuochi d'artificio in background quando si fa il gsh check.

[illegible]

## Missione n. 20:

Si richiede all'utente di utilizzare un codice di 4 lettere dopo il comando **charmiglio** in modo tale da far partire dei fuochi d'artificio. Per il completamento di questa missione è stato utilizzato il codice: **open**

[illegible]

### Missione n. 21:

Si richiede all'utente di individuare un file che contiene la parola "copper\_coin" all'interno delle directories attraverso il comando: **find iname \*coin\*** . Inoltre, una volta individuato, chiede di spostarlo nella cartella "Chest" con il comando **mv**.

```
~/Garden/Maze
[mission 21] $ find -iname *coin*
./103d0f806e35638ca4c/25a4d739aeac82/e4ff86b82/00000_copper_coin_00000

~/Garden/Maze
[mission 21] $ cd 103d0f806e35638ca4c/

~/Garden/Maze/103d0f806e35638ca4c
[mission 21] $ cd 25a4d739aeac82/

~/Garden/Maze/103d0f806e35638ca4c/25a4d739aeac82
[mission 21] $ cd e4ff86b82/

~/Garden/Maze/103d0f806e35638ca4c/25a4d739aeac82/e4ff86b82
[mission 21] $ ls
00000_copper_coin_00000

~/Garden/Maze/103d0f806e35638ca4c/25a4d739aeac82/e4ff86b82
[mission 21] $ mv 00000_copper_coin_00000 ~/Forest/Hut/Chest
mv: cannot stat '00000_copper_coin_00000': No such file or directory

~/Garden/Maze/103d0f806e35638ca4c/25a4d739aeac82/e4ff86b82
[mission 21] $ ls
00000_copper_coin_00000

~/Garden/Maze/103d0f806e35638ca4c/25a4d739aeac82/e4ff86b82
[mission 21] $ ls
00000_copper_coin_00000

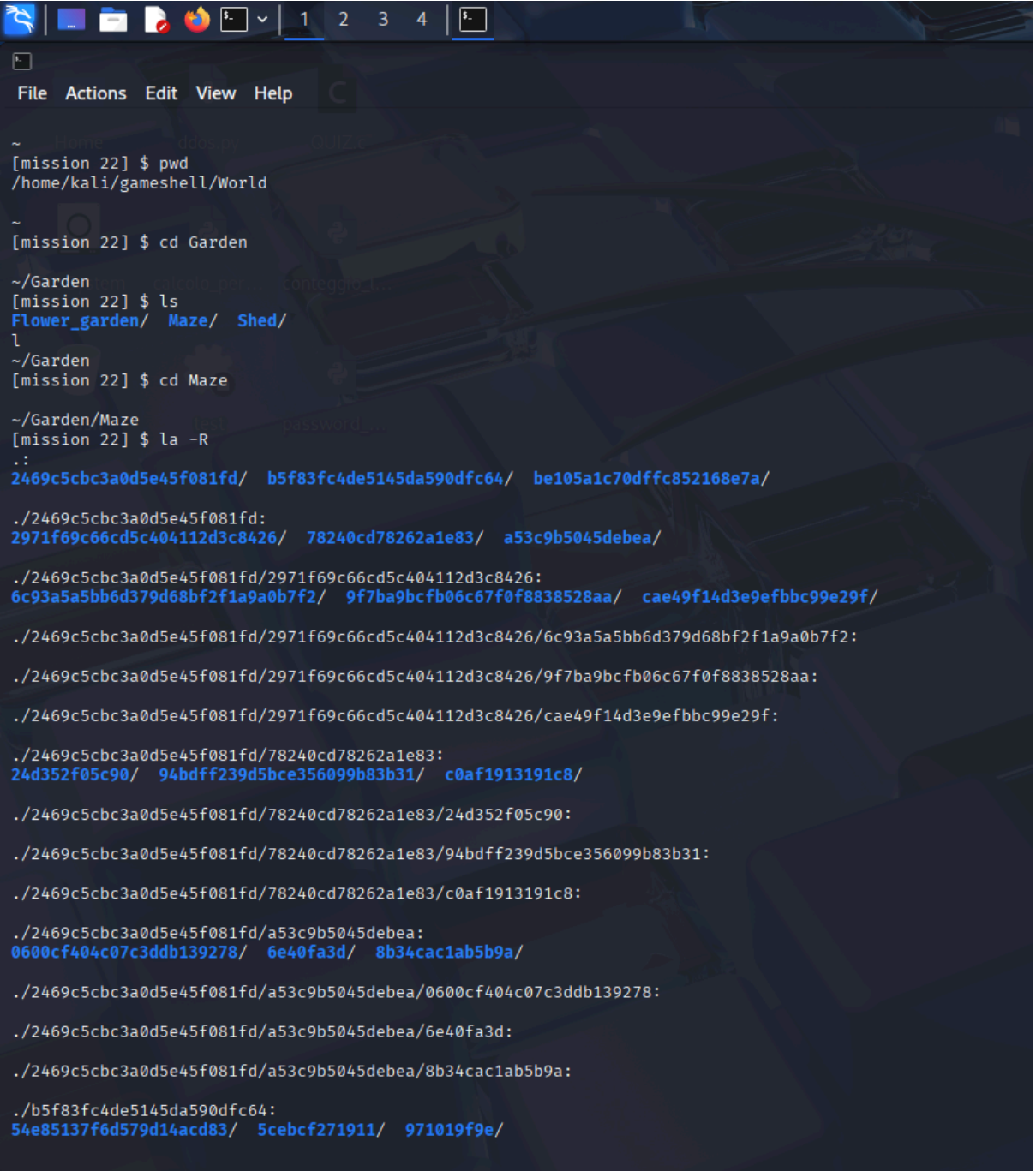
~/Garden/Maze/103d0f806e35638ca4c/25a4d739aeac82/e4ff86b82
[mission 21] $ mv 00000_copper_coin_00000 ~/Forest/Hut/Chest

~/Garden/Maze/103d0f806e35638ca4c/25a4d739aeac82/e4ff86b82
[mission 21] $ gsh check

Congratulations, mission 21 has been successfully completed!
```

## Missione n. 22:

In questo caso la missione richiede invece all'utente di spostare un file (mv) che contiene la parola "siver\_coin". Per trovare questo file si utilizza il comando **ls -R** che risulta essere lo strumento ideale per trovare un file all'interno di tutta la struttura delle directory e non solo in quella corrente.



```
[mission 22] $ pwd
/home/kali/gameshell/World

[mission 22] $ cd Garden
~/Garden
[mission 22] $ ls
Flower_garden/ Maze/ Shed/
~
~/Garden
[mission 22] $ cd Maze
~/Garden/Maze
[mission 22] $ ls -R
.:
2469c5cbc3a0d5e45f081fd/ b5f83fc4de5145da590dfc64/ be105a1c70dffc852168e7a/

./2469c5cbc3a0d5e45f081fd:
2971f69c66cd5c404112d3c8426/ 78240cd78262a1e83/ a53c9b5045debea/

./2469c5cbc3a0d5e45f081fd/2971f69c66cd5c404112d3c8426:
6c93a5a5bb6d379d68bf2f1a9a0b7f2/ 9f7ba9bcfb06c67f0f8838528aa/ cae49f14d3e9efbbc99e29f/

./2469c5cbc3a0d5e45f081fd/2971f69c66cd5c404112d3c8426/6c93a5a5bb6d379d68bf2f1a9a0b7f2:

./2469c5cbc3a0d5e45f081fd/2971f69c66cd5c404112d3c8426/9f7ba9bcfb06c67f0f8838528aa:

./2469c5cbc3a0d5e45f081fd/2971f69c66cd5c404112d3c8426/cae49f14d3e9efbbc99e29f:

./2469c5cbc3a0d5e45f081fd/78240cd78262a1e83:
24d352f05c90/ 94bdf239d5bce356099b83b31/ c0af1913191c8/

./2469c5cbc3a0d5e45f081fd/78240cd78262a1e83/24d352f05c90:

./2469c5cbc3a0d5e45f081fd/78240cd78262a1e83/94bdf239d5bce356099b83b31:

./2469c5cbc3a0d5e45f081fd/78240cd78262a1e83/c0af1913191c8:

./2469c5cbc3a0d5e45f081fd/a53c9b5045debea:
0600cf404c07c3ddb139278/ 6e40fa3d/ 8b34cac1ab5b9a/

./2469c5cbc3a0d5e45f081fd/a53c9b5045debea/0600cf404c07c3ddb139278:

./2469c5cbc3a0d5e45f081fd/a53c9b5045debea/6e40fa3d:

./2469c5cbc3a0d5e45f081fd/a53c9b5045debea/8b34cac1ab5b9a:

./b5f83fc4de5145da590dfc64:
54e85137f6d579d14acd83/ 5cebcf271911/ 971019f9e/
```

```
kali@kali: ~  
File Actions Edit View Help  
./b5f83fc4de5145da590dfc64/54e85137f6d579d14acd83:  
1141bdf4812d8aa3e84249b8780f90/ 28cebe37ca0ee3cc10ac/ f8ef7ef765f74fce35bfb4c/  
./b5f83fc4de5145da590dfc64/54e85137f6d579d14acd83/1141bdf4812d8aa3e84249b8780f90:  
./b5f83fc4de5145da590dfc64/54e85137f6d579d14acd83/28cebe37ca0ee3cc10ac:  
./b5f83fc4de5145da590dfc64/54e85137f6d579d14acd83/f8ef7ef765f74fce35bfb4c:  
./b5f83fc4de5145da590dfc64/5cebcf271911:  
1f1b2849a26042a0b572cc4694/ 82f3bdc651e48/ d54d549a2d275/  
./b5f83fc4de5145da590dfc64/5cebcf271911/1f1b2849a26042a0b572cc4694:  
./b5f83fc4de5145da590dfc64/5cebcf271911/82f3bdc651e48:  
./b5f83fc4de5145da590dfc64/5cebcf271911/d54d549a2d275:  
./b5f83fc4de5145da590dfc64/971019f9e:  
1bda1262baa90b3ed6ff3a3cc8b828a/ 5adb3090ec44db99f5/ ad1fff113e5604d6517/  
./b5f83fc4de5145da590dfc64/971019f9e/1bda1262baa90b3ed6ff3a3cc8b828a:  
./b5f83fc4de5145da590dfc64/971019f9e/5adb3090ec44db99f5:  
./b5f83fc4de5145da590dfc64/971019f9e/ad1fff113e5604d6517:  
./be105a1c70dff852168e7a:  
33a319b2efb970b9472da91cdf3dbe4/ dca2c8deb2317/ f4ef8dbb/  
./be105a1c70dff852168e7a/33a319b2efb970b9472da91cdf3dbe4:  
0fbad320d64b493029844/ 92738584473aa875e0f8/ c9640e2bd83f/  
./be105a1c70dff852168e7a/33a319b2efb970b9472da91cdf3dbe4/0fbad320d64b493029844:  
./be105a1c70dff852168e7a/33a319b2efb970b9472da91cdf3dbe4/92738584473aa875e0f8:  
./be105a1c70dff852168e7a/33a319b2efb970b9472da91cdf3dbe4/c9640e2bd83f:  
00000_silver_coin_00000  
./be105a1c70dff852168e7a/dca2c8deb2317:  
431e39f66225fbb50/ 519b67b76a40e53d1/ af33a8185b4d3d2c4/  
./be105a1c70dff852168e7a/dca2c8deb2317/431e39f66225fbb50:  
./be105a1c70dff852168e7a/dca2c8deb2317/519b67b76a40e53d1:  
./be105a1c70dff852168e7a/dca2c8deb2317/af33a8185b4d3d2c4:  
./be105a1c70dff852168e7a/f4ef8dbb:  
17bbcdc20983fd2f3e0b57427/ 9b681742327b05d7e877bbe/ dba69da3a3672dcbe2f8/  
00000_silver_coin_00000
```

```
kali@kali: ~  
File Actions Edit View Help  
./be105a1c70dff852168e7a:  
33a319b2efb970b9472da91cdf3dbe4/ dca2c8deb2317/ f4ef8dbb/  
./be105a1c70dff852168e7a/33a319b2efb970b9472da91cdf3dbe4:  
0fbad320d64b493029844/ 92738584473aa875e0f8/ c9640e2bd83f/  
./be105a1c70dff852168e7a/33a319b2efb970b9472da91cdf3dbe4/0fbad320d64b493029844:  
./be105a1c70dff852168e7a/33a319b2efb970b9472da91cdf3dbe4/92738584473aa875e0f8:  
./be105a1c70dff852168e7a/33a319b2efb970b9472da91cdf3dbe4/c9640e2bd83f:  
00000_silver_coin_00000  
./be105a1c70dff852168e7a/dca2c8deb2317:  
431e39f66225fbb50/ 519b67b76a40e53d1/ af33a8185b4d3d2c4/  
./be105a1c70dff852168e7a/dca2c8deb2317/431e39f66225fbb50:  
./be105a1c70dff852168e7a/dca2c8deb2317/519b67b76a40e53d1:  
./be105a1c70dff852168e7a/dca2c8deb2317/af33a8185b4d3d2c4:  
./be105a1c70dff852168e7a/f4ef8dbb:  
17bbcdc20983fd2f3e0b57427/ 9b681742327b05d7e877bbe/ dba69da3a3672dcbe2f8/  
./be105a1c70dff852168e7a/f4ef8dbb/17bbcdc20983fd2f3e0b57427:  
./be105a1c70dff852168e7a/f4ef8dbb/9b681742327b05d7e877bbe:  
./be105a1c70dff852168e7a/f4ef8dbb/dba69da3a3672dcbe2f8:  
~/Garden/Maze  
[mission 22] $ mv ./be105a1c70dff852168e7a/33a319b2efb970b9472da91cdf3dbe4/c9640e2bd83f/00000_silver_coin_00000 ~/Forest/Hut/Chest  
~/Garden/Maze  
[mission 22] $ gsh check  
Congratulations, mission 22 has been successfully completed!
```



## Missione n. 23:

Si richiede all'utente di spostare, attraverso il comando `mv`, i file che contengono i "gold coin". Per individuare questi file è necessario l'utilizzo del comando **find -iname** seguito dalla parola `*gold*` (tra gli asterischi così da permettere di trovare tutti i file contenenti questa parola)

```
~/Garden/Maze
[mission 23] $ find -iname *gold*
./8a56d707f9d4a6d462f6c176bc80485/be84b60d6c7e1/249834f98a567c23ff26de6ba4e/gold_coin_1
./0b7e7d9448af/1d1a51154bb804/131a576532/Gold_Coin_2

~/Garden/Maze
[mission 23] $ mv ~/Garden/Maze/./8a56d707f9d4a6d462f6c176bc80485/be84b60d6c7e1/249834f98a567c23ff26de6ba4e/gold_coin_1
mv: missing destination file operand after '/home/kali/gameshell.3/World/Garden/Maze/./8a56d707f9d4a6d462f6c176bc80485/be84b60d6c7e1/249834f98a567c23ff26de6ba4e/gold_coin_1'
Try 'mv --help' for more information.

~/Garden/Maze
[mission 23] $

~/Garden/Maze
[mission 23] $

~/Garden/Maze
[mission 23] $ mv ~/Garden/Maze/./8a56d707f9d4a6d462f6c176bc80485/be84b60d6c7e1/249834f98a567c23ff26de6ba4e/gold_coin_1 ~/Forest/Hut/Chest

~/Garden/Maze
[mission 23] $ mv ~/Garden/Maze/./0b7e7d9448af/1d1a51154bb804/131a576532/Gold_Coin_2 ~/Forest/Hut/Chest

~/Garden/Maze
[mission 23] $ gsh check

Congratulations, mission 23 has been successfully completed!
```

## Missione n.24:

Si richiede all'utente di mostrare un numero preciso di stringhe relative ad una ricetta che si trova a pagina 7 del "Book\_of\_potions", includendo il titolo e nient' altro. Per completare questa missione è necessario l'uso del comando **head**, il quale mostra le stringhe partendo dall'alto, seguito da **-n 6**, ovvero il numero preciso delle stringhe che verranno mostrate.

```
~/Garden/Maze
[mission 24] $ cd

~
[mission 24] $ cd Mountain/

~/Mountain
[mission 24] $ cd Cave/

~/Mountain/Cave
[mission 24] $ head -n6 ~/Mountain/Cave/Book_of_potions/page_07
vvvvvvvvvv
Herbal tea
^^^^^^^^^^
1) Boil water.
2) Add herbs from the forest.
3) Let it sit for five minutes and drink while hot.

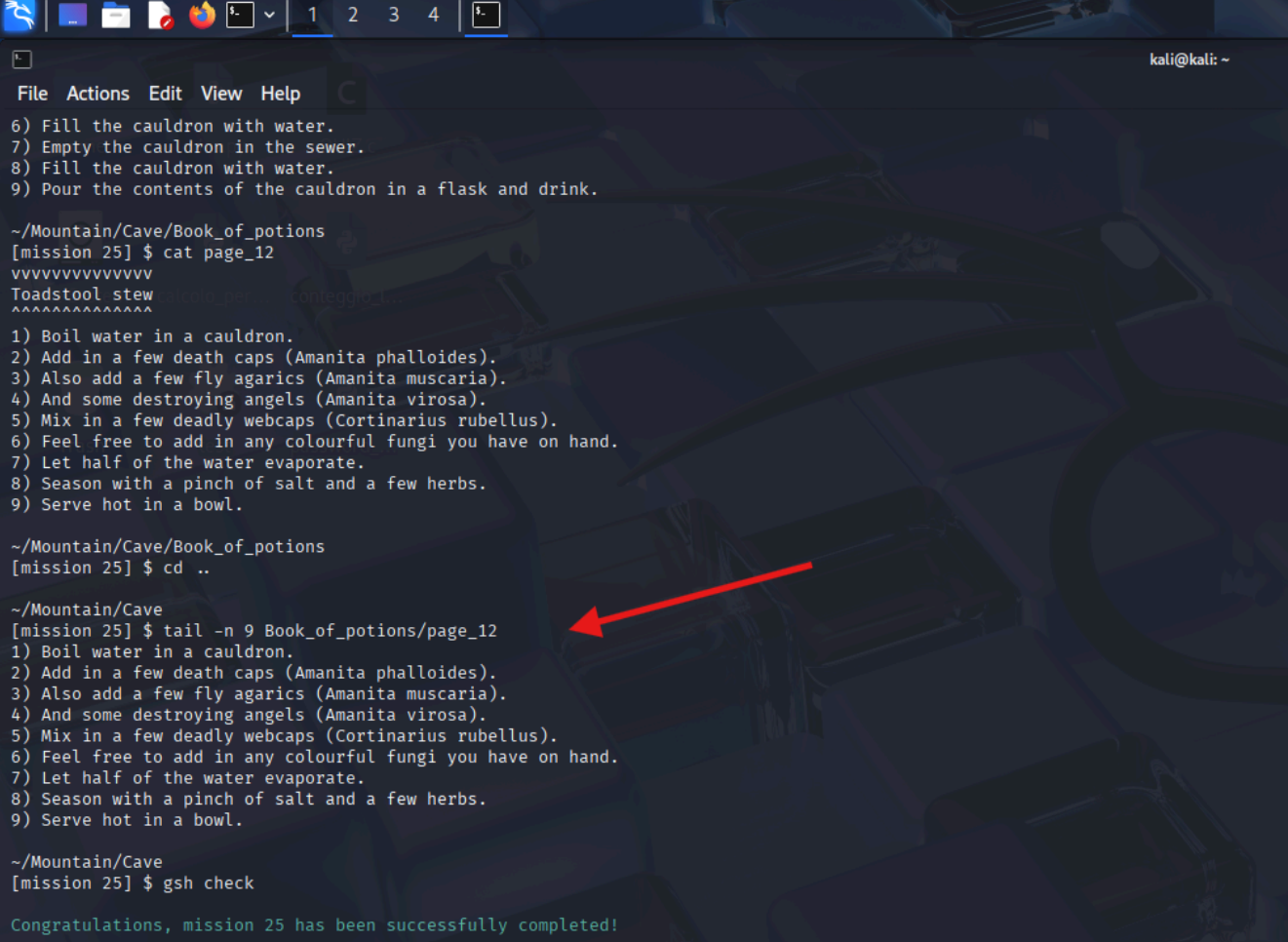
~/Mountain/Cave
[mission 24] $ gsh check

Congratulations, mission 24 has been successfully completed!
```



## Missione n. 25

Si richiede all'utente di leggere un numero preciso di stringhe finali all'interno di un documento di testo. Il comando utilizzato è **tail** seguito da **-N** (**-N** sta per il numero di stringhe da leggere partendo dal basso). In questo caso è stato utilizzato il comando **tail -9**.



```
kali@kali: ~  
File Actions Edit View Help  
6) Fill the cauldron with water.  
7) Empty the cauldron in the sewer.  
8) Fill the cauldron with water.  
9) Pour the contents of the cauldron in a flask and drink.  
~/Mountain/Cave/Book_of_potions  
[mission 25] $ cat page_12  
vvvvvvvvvvvvvvv  
Toadstool stew  
^^^^^^^^^^^^^^  
1) Boil water in a cauldron.  
2) Add in a few death caps (Amanita phalloides).  
3) Also add a few fly agarics (Amanita muscaria).  
4) And some destroying angels (Amanita virosa).  
5) Mix in a few deadly webcaps (Cortinarius rubellus).  
6) Feel free to add in any colourful fungi you have on hand.  
7) Let half of the water evaporate.  
8) Season with a pinch of salt and a few herbs.  
9) Serve hot in a bowl.  
~/Mountain/Cave/Book_of_potions  
[mission 25] $ cd ..  
~/Mountain/Cave  
[mission 25] $ tail -n 9 Book_of_potions/page_12  
1) Boil water in a cauldron.  
2) Add in a few death caps (Amanita phalloides).  
3) Also add a few fly agarics (Amanita muscaria).  
4) And some destroying angels (Amanita virosa).  
5) Mix in a few deadly webcaps (Cortinarius rubellus).  
6) Feel free to add in any colourful fungi you have on hand.  
7) Let half of the water evaporate.  
8) Season with a pinch of salt and a few herbs.  
9) Serve hot in a bowl.  
~/Mountain/Cave  
[mission 25] $ gsh check  
Congratulations, mission 25 has been successfully completed!
```

## Missione n. 26:

Si richiede all'utente di stampare, tramite l'uso del comando **cat**, 2 file che si completano.

```
~/Mountain/Cave
[mission 26] $ cat ~/Mountain/Cave/Book_of_potions/page_01 cat ~/Mountain/Cave/Book_of_potions/page_02
vvvvvvvvvvvvvvvvvvvvvvvvvvvvvv
Transformation potion
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
1) Boil water in a cauldron.
2) Add 3 measures of fluxweed to the cauldron.
3) Add 2 bundles of knotgrass to the cauldron.
4) Stir 4 times, clockwise.
5) Wave your wand then let potion brew for 80 minutes.
6) Add 4 leeches to the cauldron.
7) Crush 2 scoops of lacewing flies to a fine paste.
8) Add 2 measures of the crushed lacewings to the cauldron.
9) Heat for 30 seconds on a low heat.
cat: cat: No such file or directory
10) Add 3 measures of boomslang skin to the cauldron.
11) Crush a bicorn horn into a fine powder.
12) Add 1 measure of the crushed horn to the cauldron.
13) Heat for 20 seconds at a high temperature.
14) Wave your wand then let potion brew for 24 hours.
15) Add 1 additional scoop of lacewings to the cauldron.
16) Stir 3 times, counter-clockwise.
17) Split potion into multiple doses, if desired.
18) Add a pieces of the person you wish to become.
19) Wave your wand to complete the potion.

~/Mountain/Cave
[mission 26] $ gsh check
cat: cat: No such file or directory

Congratulations, mission 26 has been successfully completed!
```

## Missione n. 27:

Si richiede all'utente di stampare, di due documenti diversi, le ultime 16 stringhe. Qui i comandi necessari al completamento della richiesta sono **cat**, per la stampa del contenuto del documento e **tail -16**, per stampare le ultime 16 stringhe.

```
~/Mountain/Cave
[mission 27] $ cat Book_of_potions/page_03 Book_of_potions/page_04 | tail -n 16
1) Fill a cauldron with used bath water.
2) Put a moderately large frog in the water.
3) Let the preparation rest overnight.
4) The next morning thank and free your little green friend.
5) Boil the water and add in a few sticks of oak tree.
6) Crush 5 river stones to a fine powder.
7) Mix in a third of the powder and stir vigorously.
8) Let the preparation rest for a day.
9) Add hairs from the tail of a squirrel (willingly given).
10) Add the remaining stone powder.
11) Stir the potion very vigorously, in all directions.
12) Take some time to rest after such an effort.
13) Rest a little bit more.
14) Even take a nap if you want.
15) Add a few larch tree needles for seasoning.
16) Drink the potion from the cauldron.

~/Mountain/Cave
[mission 27] $ gsh check

Congratulations, mission 27 has been successfully completed!

[ progress was saved in /home/kali/gameshell-save.sh ]

|
--+-----+--
```

## Missione n. 28:

Si richiede all'utente di stampare delle stringhe di testo precise, in particolare quelle centrali. In questo caso è necessario utilizzare i comandi **head** e **tail**. Più precisamente, **head -n6** e **tail -n3**.

```
~/Mountain/Cave
[mission 28] $ cat Book_of_potions/page_13 | head -n 6 | tail -n 3
1) Boil water in a big pot.
2) Condense the vapor in a fresh container.
3) Add minerals for a better taste (optional).

~/Mountain/Cave
[mission 28] $ gsh check

Congratulations, mission 28 has been successfully completed!
```

## Missione n. 29:

Si richiede all'utente di individuare, tramite il comando **ps**, i processi in esecuzione e i relativi PID, al fine di porre fine al processo "spell". Una volta individuato, tramite il comando **kill**, seguito dal **PID**, si può completare la missione.

```
~/Mountain/Cave
[mission 29] $ ps
  PID TTY          TIME CMD
 23621 pts/0    00:00:00 zsh
 23674 pts/0    00:00:00 bash
 23731 pts/0    00:00:00 bash
 56707 pts/0    00:00:00 spell
 57029 pts/0    00:00:00 ps

~/Mountain/Cave
[mission 29] $
    *#@*
    0_**/~
    !$-#

kill
    *#@*
    0_**/~
    !$-#

    *#@*
    0_**/~
    !$-#

kill 56707
bash: kill: kill: arguments must be process or job IDs

~/Mountain/Cave
[mission 29] $ gsh check

Congratulations, mission 29 has been successfully completed!
```

## Missione n. 30:

Si richiede all'utente di terminare dei processi in esecuzione per i quali il comando **kill** sembra non bastare. In questo caso, per porre fine a questi processi, è necessario l'utilizzo del comando **kill -9**, seguito dal/dai **PID**, il quale termina forzatamente i relativi processi.

```
~/Mountain/Cave
[mission 30] $ ps
  PID TTY          TIME CMD
 23621 pts/0    00:00:00 zsh
 23674 pts/0    00:00:00 bash
 23731 pts/0    00:00:00 bash
 65847 pts/0    00:00:00 spell
 66216 pts/0    00:00:00 spell
 66279 pts/0    00:00:00 ps

~/Mountain/Cave
[mission 30] $

^C

~/Mountain/Cave
[mission 30] $ kill -9

65847

~/Mountain/Cave
[mission 30] $ kill -9 6621

6

~/Mountain/Cave
[mission 30] $ gsh check

Congratulations, mission 30 has been successfully completed!
```

## Missione n. 31:

Si richiede all'utente di eliminare determinati "spell" e di lasciare gli altri. Per identificare tutti i PID dei processi appartenenti ad uno specifico è necessario utilizzare il comando **ps** seguito dal PID del processo. Una volta identificati, per eliminarli, è necessario quindi utilizzare il comando **kill**, seguito da tutti i PID trovati.

```
File  Actions  Edit  View  Help
461880 pts/1    00:00:00 zsh
463307 pts/1    00:00:00 bash
463363 pts/1    00:00:00 bash
470637 pts/1    00:00:00 sleep
470671 pts/1    00:00:00 nice_fairy
470672 pts/1    00:00:00 mischievous_imp
470678 pts/1    00:00:00 spell
470679 pts/1    00:00:00 spell
470680 pts/1    00:00:00 spell
470681 pts/1    00:00:00 tail
470693 pts/1    00:00:00 spell
470694 pts/1    00:00:00 spell
470695 pts/1    00:00:00 spell
470696 pts/1    00:00:00 tail
475688 pts/1    00:00:00 sleep
475689 pts/1    00:00:00 sleep
475768 pts/1    00:00:00 sleep
475769 pts/1    00:00:00 sleep
475848 pts/1    00:00:00 sleep
475849 pts/1    00:00:00 sleep
475850 pts/1    00:00:00 ps

~/Castle/Cellar
[mission 31] $ pstree -p 470672
mischievous_imp(470672)─spell(470693)─sleep(477326)
                        └─spell(470694)─sleep(477391)
                        └─spell(470695)─sleep(477494)
                        └─tail(470696)

~/Castle/Cellar
[mission 31] $ kill 470693 470694 470695

~/Castle/Cellar
[mission 31] $ rm -rf *coal*

~/Castle/Cellar
[mission 31] $ gsh check

Congratulations, mission 31 has been successfully completed!
```



## Missione n. 32:

Si richiede all'utente di effettuare dei semplici calcoli matematici.

```
~/Castle/Cellar
[mission 32] $ gsh goal

/σ\ \_||
( \_ \_||
\_/ \_/||

Mission goal

To get better in the magical art, one needs to know mental math.

Get ready, because Merlin is about to test your precision with sums.

Run the command ``gsh check`` to start.

/σ\ \_||
( \_ \_||
\_/ \_/||

~/Castle/Cellar
[mission 32] $ gsh check
8 + 30 = ?? 38
76 + 44 = ?? 120
56 + 53 = ?? 109
70 + 13 = ?? 83
81 + 7 = ?? 88

Congratulations, mission 32 has been successfully completed!
```

## Missione n. 33:

Si richiede all'utente di effettuare dei calcoli matematici entro un determinato limite di tempo. Il tempo a disposizione rende impossibile il completamento di questa richiesta e per questo è necessario utilizzare il comando `<` seguito dal file "Mathematics\_101" ( un file che contiene tutti i calcoli e i relativi risultati richiesti) in modo tale che questo sostituisca l'input da tastiera. Dunque, il comando necessari al completamento della missione è il seguente: **`gsh check < Mathematics_101`** .

```
~
[mission 33] $ gsh check < Mathematics_101
bash: Mathematics_101: No such file or directory

~
[mission 33] $ pwd
/home/kali/gameshell.1/World

~
[mission 33] $ cd Castle

~/Castle
[mission 33] $ cd Main_building

~/Castle/Main_building
[mission 33] $ cd Library

~/Castle/Main_building/Library
[mission 33] $ gsh check < Mathematics_101
41 * 93 = ?? 38 * 16 = ?? 45 * 60 = ?? 91 * 15 = ?? 48 * 29 = ?? 61 * 33 = ?? 18 * 35 = ?? 43 * 53 = ?? 53 * 8 = ?? 33 * 56 = ?? 56 * 27
* 14 = ?? 69 * 56 = ?? 57 * 58 = ?? 26 * 29 = ?? 65 * 39 = ?? 95 * 26 = ?? 55 * 53 = ?? 19 * 1 = ?? 58 * 51 = ?? 3 * 52 = ?? 30 * 73 = ?
?? 55 * 69 = ?? 19 * 33 = ?? 90 * 49 = ?? 93 * 54 = ?? 96 * 2 = ?? 37 * 98 = ?? 87 * 52 = ?? 80 * 31 = ?? 91 * 53 = ?? 9 * 27 = ?? 25 *
* 99 = ?? 51 * 70 = ?? 78 * 74 = ?? 32 * 26 = ?? 53 * 17 = ?? 12 * 30 = ?? 80 * 94 = ?? 60 * 45 = ?? 32 * 98 = ?? 26 * 78 = ?? 1 * 7 =
4 = ?? 42 * 61 = ?? 59 * 78 = ?? 15 * 9 = ?? 61 * 93 = ?? 84 * 76 = ?? 94 * 10 = ?? 45 * 50 = ?? 77 * 59 = ?? 62 * 58 = ?? 55 * 83 = ??
65 * 34 = ?? 10 * 15 = ?? 51 * 22 = ?? 39 * 82 = ?? 90 * 65 = ?? 71 * 9 = ?? 54 * 50 = ?? 63 * 76 = ??

Congratulations, mission 33 has been successfully completed!
```

## Missione n. 34:

Si richiede all'utente di salvare un elenco di libri di incantesimi all'interno di un file chiamato "inventory.txt" che si trova all'interno della cartella Drawer. Per compiere questa richiesta è necessario individuare tutti i file contenenti la parola grimoire, utilizzando l'asterisco (\*) alla fine della parola, seguita dal comando › , a sua volta seguito dal percorso dove si vuole salvare questi file.

```
~/Castle/Main_building
[mission 34] $ ls
Library/ Throne_room/

~/Castle/Main_building
[mission 34] $ cd Library/

~/Castle/Main_building/Library
[mission 34] $ cd M
Mathematics_101 Merlin_s_office/

~/Castle/Main_building/Library
[mission 34] $ cd Merlin_s_office/

~/Castle/Main_building/Library/Merlin_s_office
[mission 34] $ ls
candle      grimoire_11253 grimoire_13655 grimoire_15539 grimoire_18082 grimoire_21273 grimoire_2360 grimoire_25184 grimoire_26790 grimoire_29388 grimoire_31507 grimoire_5820 grimoire_7768
Drawer/     grimoire_11530 grimoire_13971 grimoire_16079 grimoire_18250 grimoire_22023 grimoire_2385 grimoire_25607 grimoire_27074 grimoire_29783 grimoire_31850 grimoire_5883 grimoire_7978
grimoire_10116 grimoire_11618 grimoire_14316 grimoire_16214 grimoire_19027 grimoire_22306 grimoire_24254 grimoire_25769 grimoire_27386 grimoire_29906 grimoire_31855 grimoire_6621 grimoire_8451
grimoire_10606 grimoire_12147 grimoire_1460 grimoire_16505 grimoire_20411 grimoire_22793 grimoire_24618 grimoire_25916 grimoire_27683 grimoire_30087 grimoire_32552 grimoire_6704 grimoire_9129
grimoire_10714 grimoire_12211 grimoire_14624 grimoire_16562 grimoire_20586 grimoire_23076 grimoire_24783 grimoire_26087 grimoire_27920 grimoire_30367 grimoire_363 grimoire_6804 grimoire_9150
grimoire_1072 grimoire_12533 grimoire_14759 grimoire_16828 grimoire_20887 grimoire_23127 grimoire_24808 grimoire_26282 grimoire_28516 grimoire_30453 grimoire_3987 grimoire_6842 grimoire_9379
grimoire_11105 grimoire_13026 grimoire_14944 grimoire_1719 grimoire_21115 grimoire_23169 grimoire_24904 grimoire_26361 grimoire_29 grimoire_30552 grimoire_4563 grimoire_7004 grimoire_9379
grimoire_11187 grimoire_13178 grimoire_15504 grimoire_18033 grimoire_21219 grimoire_23283 grimoire_2495 grimoire_26391 grimoire_2907 grimoire_31360 grimoire_5422 grimoire_7125

~/Castle/Main_building/Library/Merlin_s_office
[mission 34] $ ls grimoire* > Drawer/inventory.txt
~/Castle/Main_building/Library/Merlin_s_office
[mission 34] $ gsh check

Congratulations, mission 34 has been successfully completed!
```

## Missione n. 35:

Si richiede di mostrare i nomi di alcuni libri ( grimoire ) che contengono la stringa “gsh”, utilizzando un comando che non tenga conto delle maiuscole e minuscole e che bypassi i messaggi di errore che si possono riscontrare nell’ apertura di questi file. Necessario dunque utilizzare i comandi **grep**, seguito da “gsh”, per individuare i file contenenti questa parola, **-i -l** per individuare i file indipendentemente dalle maiuscole e minuscole, e infine il comando **/dev/null** per eliminare i messaggi di errore generati all’apertura di questi file.

```
/b\
Mission goal
Merlin's old alchemy books are kept in his office, in the library. You need to output a list of all the books containing the alchemical compound 'gsh' (for "Glutathione").
Beware, it can be spelled with a mix of letters in upper and lowercase.

Some of those books are locked, and you are not allowed to consult them. Ignore them.

Remark

Your last command should display the list of books mentioning 'gsh', and nothing else.
In particular, you should not show the chemical compounds themselves and no error message should appear.

Useful commands

grep STRING FILE1 ... FILEn
Search for a string inside a file and print the corresponding lines.
Some interesting options are
-l : do not print the lines, but only the filenames containing
the string
-i : do not differentiate between uppercase and lowercase

COMMAND > FILE
Send messages (stdout) to the file.
(The file is overwritten.)

COMMAND 2> FILE
Send error messages (stderr) to the file.
(The file is overwritten.)

/dev/null
"Virtual" file that is universally empty: anything sent to it disappears.

Explanations

Every command can output messages to the screen in two ways:
- standard printing ("stdout")
- error printing ("stderr")
By default, both kind of messages are shown on the screen.

It is however possible to redirect both of them to files independently.
/b\
```

```
~/Castle/Main_building/Library/Merlin_s_office
[mission 35] $ grep "gsh" -i -l *grimoire* 2> /dev/null
grimoire_aeTSxiMLhXq
grimoire_aOwfVYnhBxdMQr
grimoire_BqzHWZGifssWkDPEavEr
grimoire_BzobRTDsdyoXeiBzUUcuBsFH
grimoire_cCbKsNdZoy
grimoire_ccQYeWxYJRapawSdDFRPeEuIMknj
grimoire_CeiAOZAcjHtzOEcsvBFDSSWlwILYX
grimoire_CpmUvySpZix
grimoire_dJBctKKAovBrW
grimoire_eGACeSiqHEizAdmOtikhFY
grimoire_HUKijVvqEObbatIJYfuWBmt
grimoire_lbOTwRXq
grimoire_LJMEplrdOBXMFncWDxObk
grimoire_mIIfsHJuJAKQpUyjmXZkCFQ
grimoire_MjppEdsEkzGzwmXcFNAOEKZhRpeeci
grimoire_mkBELDsQlVrLl
grimoire_NKGAHdDbcnCpJC
grimoire_NLnuHVBMBEzqWZdiPENNIvIWlJe
grimoire_oaKjuQNjDjbZBe
grimoire_oMdpmnSaVqdcrcVCLQyEcmeWEeYU
grimoire_PUrGbJGEKMQXIBeNdNSgajC
grimoire_RCgunwxm
grimoire_rIKwXBVjTQHBoclHgZwJF
grimoire_rxnzImMGeqSjg
grimoire_sCixTaizIBrBECKyfhgsGcB
grimoire_SgvcjrDTfQgFuFn
grimoire_skutgWOSQqTawXy
grimoire_vmcPzxpOuA
grimoire_WPYkEoEckDYstxseXVyUQXApRyBskwL
grimoire_xHEJzcFFcIKHxRXMxsjnd
grimoire_XLMLSwTssFFCxfGWejuCwGYCXIj
grimoire_ZfwEvZWCFehocxmtorwXZJNySbCN

~/Castle/Main_building/Library/Merlin_s_office
[mission 35] $ gsh check

Congratulations, mission 35 has been successfully completed!
```

## Missione n. 36:

Si richiede all'utente di individuare e fornire una "secret key" che si trova all'interno dell'eseguibile "Merlin". Questa key contiene degli errori che l'utente è chiamato a risolvere. Per prima cosa è necessario avviare Merlin con il comando: **./Merlin** seguito dal comando **/dev/null** (per eliminare gli errori che potrebbero generarsi). Successivamente si deve procedere a salvare la key all'interno di un file chiamato "key.txt" in modo tale da completare la missione attraverso il comando: **gsh check < key.txt**.

```

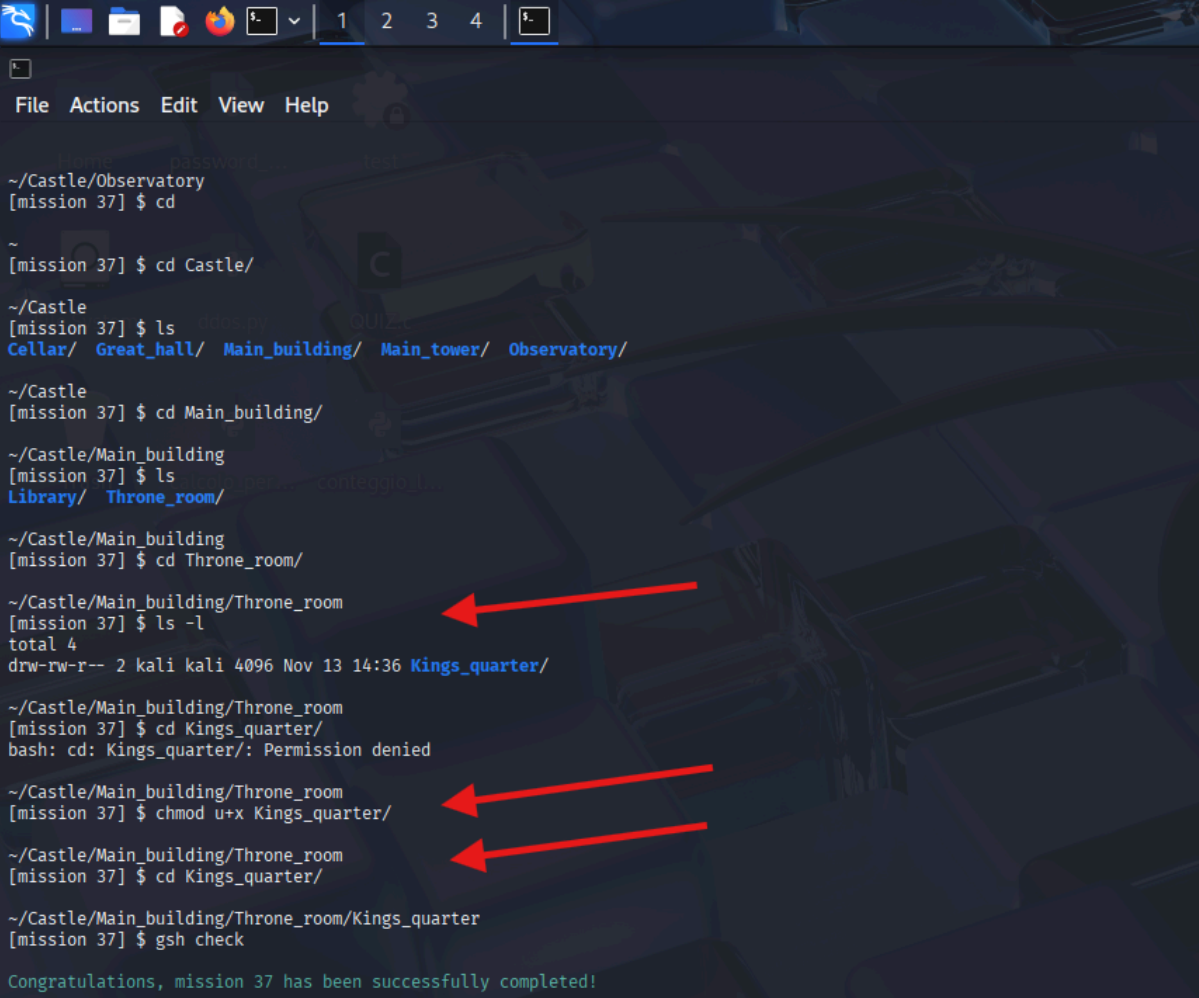
Mission goal
Merlin has turned crazy... He paces around the observatory tower and mumbles incoherently.
You need to filter out his ramblings to discover the secret key he is the only one to know.
Hint
It is likely that Merlin's message is riddled with errors.
To complete this mission, you have to give the secret key **using a file redirection**.
Useful commands
./FILE
Run an executable file as a program. Needless to say that Merlin is executable...
COMMAND > FILE
Send standard output (stdout) for the command to a file.
(The file is overwritten.)
COMMAND 2> FILE
Send error output (stderr) for the command to a file.
(The file is overwritten.)
COMMAND < FILE
Send the file to the standard input (stdin) of the command.
```

```

File Actions Edit View Help
~/Castle/Main_building/Library/Merlin_s_office
[mission 36] $ cd Castle/Observatory
bash: cd: Castle/Observatory: No such file or directory
~/Castle/Main_building/Library/Merlin_s_office
[mission 36] $ cd
~/Castle
[mission 36] $ cd Castle
~/Castle
[mission 36] $ cd Observatory/
~/Castle/Observatory
[mission 36] $ ls
merlin star_chart
~/Castle/Observatory
[mission 36] $ ./Merlin 2> /dev/null
~/Castle/Observatory
[mission 36] $ ./merlin 2> /dev/null
THESECREKEYISONSTDERR
~/Castle/Observatory
[mission 36] $ ./merlin
ifYCeXtSiGxqPInxmMhNIUNHnzpIhdEnlklQdzWk8TF5RDGKjIddLuFECnbaXfKcKcBCgzSkFakPHAF1UaTRhfjxxiJnpwXwysGfDWYAOKhETGvKbmPyjmmKujKEISgMDiyKolyjvIzZSFiyKpYaqF1OLEB1wbSGUoJiGvETNfNOqxxSQRcfzcoTDLEJCKrKRDYDKfgaztxLPztpUCeZCQtpSR
~/Castle/Observatory
[mission 36] $ ./merlin 2> key.txt
THESECREKEYISONSTDERR
~/Castle/Observatory
[mission 36] $ ls
key.txt merlin star_chart
~/Castle/Observatory
[mission 36] $ gsh check < key.txt
What is the secret key?
Congratulations, mission 36 has been successfully completed!
```

## Missione n. 37:

Si richiede all'utente di accedere alla cartella "Kings\_quarter" che si trova nella "Throne\_room". L'accesso a questa directory risulta negato poiché i permessi, visibili con il comando **ls -l** risultano essere **rw-rw-r**, ovvero non eseguibile dall'utente. Per il completamento di questa missione è dunque necessario cambiare i permessi con il comando **chmod**. Il comando che bisogna utilizzare è il seguente: **chmod u+x nome\_del\_file**.



```
~/Castle/Observatory
[mission 37] $ cd

~
[mission 37] $ cd Castle/

~/Castle
[mission 37] $ ls
Cellar/ Great_hall/ Main_building/ Main_tower/ Observatory/

~/Castle
[mission 37] $ cd Main_building/

~/Castle/Main_building
[mission 37] $ ls
Library/ Throne_room/

~/Castle/Main_building
[mission 37] $ cd Throne_room/

~/Castle/Main_building/Throne_room
[mission 37] $ ls -l
total 4
drw-rw-r-- 2 kali kali 4096 Nov 13 14:36 Kings_quarter/

~/Castle/Main_building/Throne_room
[mission 37] $ cd Kings_quarter/
bash: cd: Kings_quarter/: Permission denied

~/Castle/Main_building/Throne_room
[mission 37] $ chmod u+x Kings_quarter/

~/Castle/Main_building/Throne_room
[mission 37] $ cd Kings_quarter/

~/Castle/Main_building/Throne_room/Kings_quarter
[mission 37] $ gsh check

Congratulations, mission 37 has been successfully completed!
```



### Missione n. 38:

Si richiede all'utente di leggere una "secret\_note" all'interno del "Kings\_quarter". Qui occorre innanzitutto individuare il file con il comando **ls (ls -A)**, visualizzare i permessi con il comando **ls -l** e successivamente cambiare i permessi del file interessato. In questo caso il file non aveva i permessi di lettura e quindi è stato necessario utilizzare il seguente comando: **chmod u+r .secret\_note** per renderla leggibile dall'utente. Una volta aggiunto il permesso di lettura è possibile leggere il contenuto con il comando **cat**.

```

File Actions Edit View Help

Mission goal
The King is rumored to keep the combination to his safe written on a note in his room.
Find that combination.

Remark
The King probably tried to make this note unreadable!

Useful commands
ls -l
Lists the files with their important meta-data.
Access permissions are listed first on each line.
chmod [OPTIONS] FILE
Modifies the permissions for the file (or directory).
Refer to the manual to discover which options are available.

~/Castle/Main_building/Throne_room/Kings_quarter
[mission 38] $ la
note .secret_note

~/Castle/Main_building/Throne_room/Kings_quarter
[mission 38] $ la -l
total 8
-rw-rw-r-- 1 kali kali 11 Nov 18 13:42 note
--w--w-- 1 kali kali 11 Nov 18 13:42 .secret_note

~/Castle/Main_building/Throne_room/Kings_quarter
[mission 38] $ chmod u+r .secret_note

~/Castle/Main_building/Throne_room/Kings_quarter
[mission 38] $ cat .secret_note
6054620351

~/Castle/Main_building/Throne_room/Kings_quarter
[mission 38] $ gsh check
What's the combination to open the King's safe? 6054620351

Congratulations, mission 38 has been successfully completed!

```

### **Missione n. 39:**

Si richiede all'utente di leggere un file "crown" all'interno di un file nascosto "Safe" e di spostarlo nella cartella "Chest". Per prima cosa occorre individuare il file con il comando **la (ls -A)**, successivamente con il comando **chmod**, occorre cambiare i permessi di questo file in modo tale da leggere il contenuto. Una volta aperto il file, con il comando **mv**, si può procedere allo spostamento nella cartella indicata.

```
File Actions Edit View Help
~
~/Castle/Main_building/Throne_room
[mission 39] $ la
Kings_quarter/ Safe/
~/Castle/Main_building/Throne_room
[mission 39] $ cd Safe/
bash: cd: Safe/: Permission denied
~/Castle/Main_building/Throne_room
[mission 39] $ chmod u+rwX Safe/
~/Castle/Main_building/Throne_room
[mission 39] $ cd Safe/
~/Castle/Main_building/Throne_room/Safe
[mission 39] $ la
crown
~/Castle/Main_building/Throne_room/Safe
[mission 39] $ cat crown
cat: crown: Permission denied
~/Castle/Main_building/Throne_room/Safe
[mission 39] $ chmod u+rwX crown
~/Castle/Main_building/Throne_room/Safe
[mission 39] $ cat crown
^-+.^-^
(^V^V^V^
\@*@\@/
{ _927_ }
~/Castle/Main_building/Throne_room/Safe
[mission 39] $ mv crown ~Forest/Hut/Chest/
mv: cannot move 'crown' to '~Forest/Hut/Chest/': No such file or directory
~/Castle/Main_building/Throne_room/Safe
[mission 39] $ mv crown ~/Forest/Hut/Chest
~/Castle/Main_building/Throne_room/Safe
[mission 39] $ gsh check
What are the 3 digits inscribed on the base of the crown? 927
Congratulations, mission 39 has been successfully completed!
```

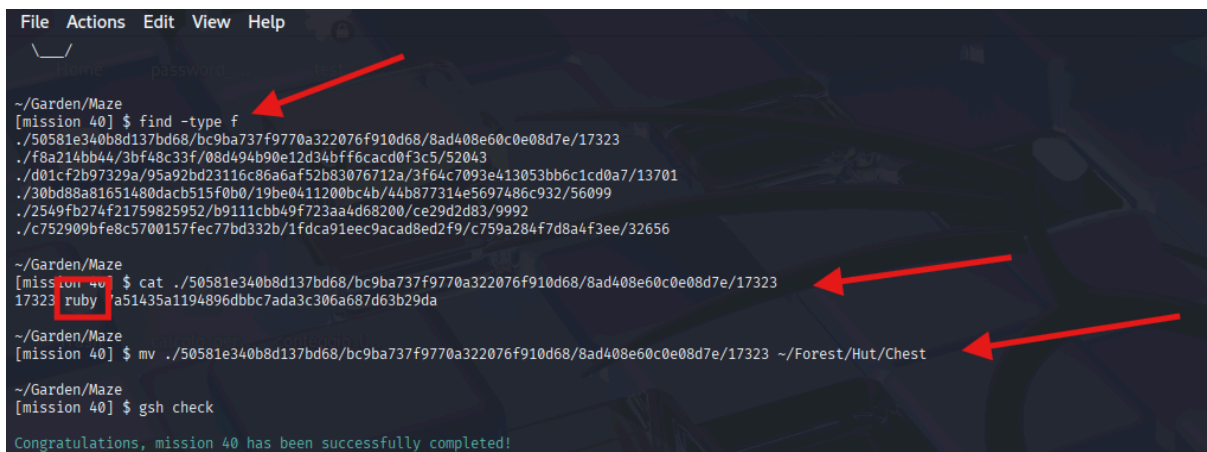
## Missione n. 40:

Si richiede all'utente di individuare un file che contiene la parola "ruby" e successivamente di spostarlo nella cartella "Chest".

E' necessario, al fine di individuare il file, il comando **find -type f**, il quale ci mostra tutti i file all'interno di una determinata directory.

Con il comando **cat** si puo' procedere all'apertura di ogni file sino all'identificazione di quello corretto.

Una volta individuato il file si puo' procedere allo spostamento con il comando **mv**.

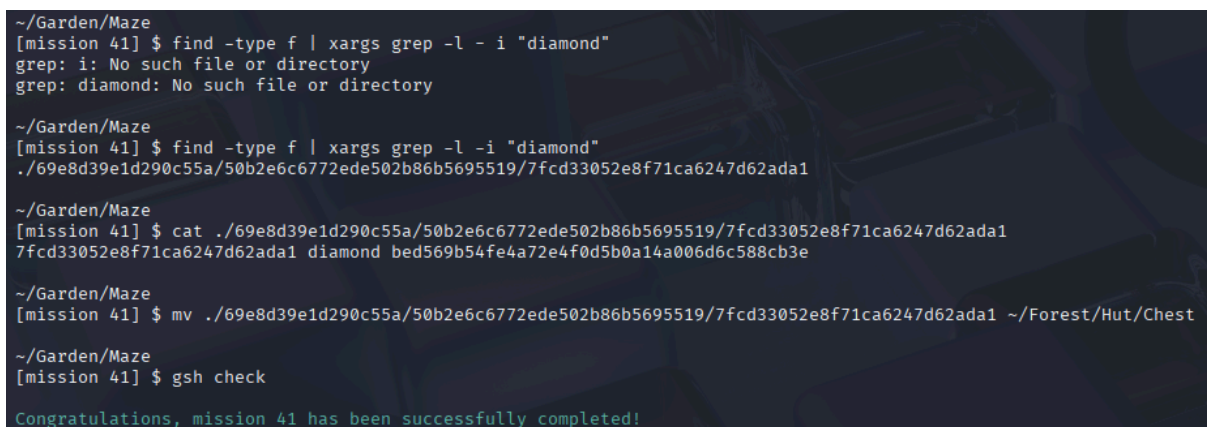
A terminal window with a dark background and a file explorer interface in the background. The terminal shows the following commands and output:

```
~/Garden/Maze  
[mission 40] $ find -type f  
./50581e340b8d137bd68/bc9ba737f9770a322076f910d68/8ad408e60c0e08d7e/17323  
./f8a214bb44/3bf48c33f/08d494b90e12d34bff6cadc0f3c5/52043  
./d01cf2b97329a/95a92bd23116c86a6af52b83076712a/3f64c7093e413053bb6c1cd0a7/13701  
./30bd88a81651480dacb515f0b0/19be0411200bc4b/44b877314e5697486c932/56099  
./2549fb274f21759825952/b9111cbb49f723aa4d68200/ce29d2d83/9992  
./c752909bfe8c5700157fec77bd332b/1fdca91eec9acad8ed2f9/c759a284f7d8a4f3ee/32656  
  
~/Garden/Maze  
[mission 40] $ cat ./50581e340b8d137bd68/bc9ba737f9770a322076f910d68/8ad408e60c0e08d7e/17323  
17323 ruby fa51435a1194896dbbc7ada3c306a687d63b29da  
  
~/Garden/Maze  
[mission 40] $ mv ./50581e340b8d137bd68/bc9ba737f9770a322076f910d68/8ad408e60c0e08d7e/17323 ~/Forest/Hut/Chest  
  
~/Garden/Maze  
[mission 40] $ gsh check  
  
Congratulations, mission 40 has been successfully completed!
```

Three red arrows point to the 'find' command, the file path in the 'cat' command, and the destination path in the 'mv' command.

## Missione n. 41:

Si richiede all'utente di spostare un file, all'interno della cartella "Maze", che contiene la parola "diamond". Qui è necessario l'utilizzo dei comandi: **find -type f**, per la ricerca nella directory, **xargs grep -l -i "diamond"**, per la ricerca dei file contenenti la parola diamond, indipendentemente dalla presenza di lettere maiuscole e minuscole.

A terminal window with a dark background and a file explorer interface in the background. The terminal shows the following commands and output:

```
~/Garden/Maze  
[mission 41] $ find -type f | xargs grep -l -i "diamond"  
grep: i: No such file or directory  
grep: diamond: No such file or directory  
  
~/Garden/Maze  
[mission 41] $ find -type f | xargs grep -l -i "diamond"  
./69e8d39e1d290c55a/50b2e6c6772ede502b86b5695519/7fcd33052e8f71ca6247d62ada1  
  
~/Garden/Maze  
[mission 41] $ cat ./69e8d39e1d290c55a/50b2e6c6772ede502b86b5695519/7fcd33052e8f71ca6247d62ada1  
7fcd33052e8f71ca6247d62ada1 diamond bed569b54fe4a72e4f0d5b0a14a006d6c588cb3e  
  
~/Garden/Maze  
[mission 41] $ mv ./69e8d39e1d290c55a/50b2e6c6772ede502b86b5695519/7fcd33052e8f71ca6247d62ada1 ~/Forest/Hut/Chest  
  
~/Garden/Maze  
[mission 41] $ gsh check  
  
Congratulations, mission 41 has been successfully completed!
```

## Missione n. 42:

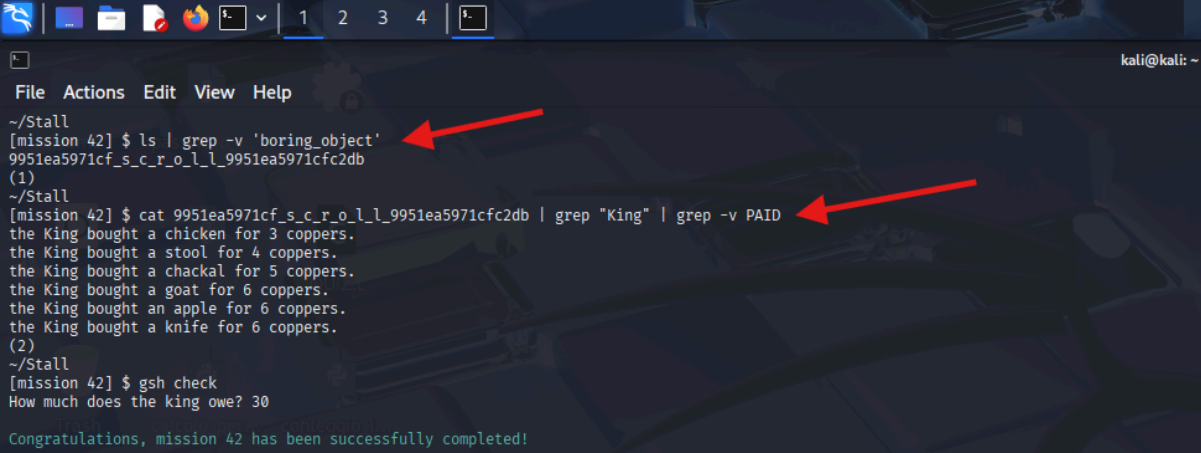
Si richiede all'utente di indicare il debito del "King" andando ad analizzare un file attraverso solamente 3 comandi all'interno della cartella "Stall".

E' necessario qui identificare il file corretto tramite il filtraggio di quelli presenti nella cartella attraverso il comando **ls | grep "boring\_object"**.

Identificato il file, bisogna individuare al suo interno tutti gli oggetti acquistati dal "King" che pero' non contengono la parola "PAID", tramite il comando: **cat <file> | grep "King" | grep -v PAID**.

Una volta aperto il file, vengono mostrati gli oggetti che il Re aveva acquistato.

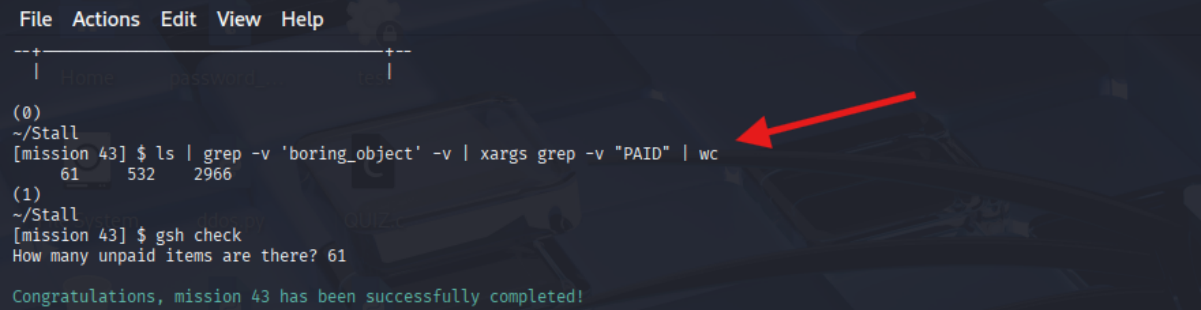
Sommando i relativi costi risulta che il Re è in debito di 30 coppers.



```
File Actions Edit View Help
~/Stall
[mission 42] $ ls | grep -v 'boring_object'
9951ea5971cf_s_c_r_o_l_l_9951ea5971cfc2db
(1)
~/Stall
[mission 42] $ cat 9951ea5971cf_s_c_r_o_l_l_9951ea5971cfc2db | grep "King" | grep -v PAID
the King bought a chicken for 3 coppers.
the King bought a stool for 4 coppers.
the King bought a chackal for 5 coppers.
the King bought a goat for 6 coppers.
the King bought an apple for 6 coppers.
the King bought a knife for 6 coppers.
(2)
~/Stall
[mission 42] $ gsh check
How much does the king owe? 30
Congratulations, mission 42 has been successfully completed!
```

## Missione n. 43:

Si richiede all'utente di individuare il numero di oggetti nella stalla che risultano "UNPAID" attraverso un unico comando. E' necessario in questo caso escludere gli oggetti che contengono le parole "boring\_object" e la parola "PAID" attraverso il comando: **ls | grep -v 'boring\_object' -v | xargs grep -v "PAID" | wc**.

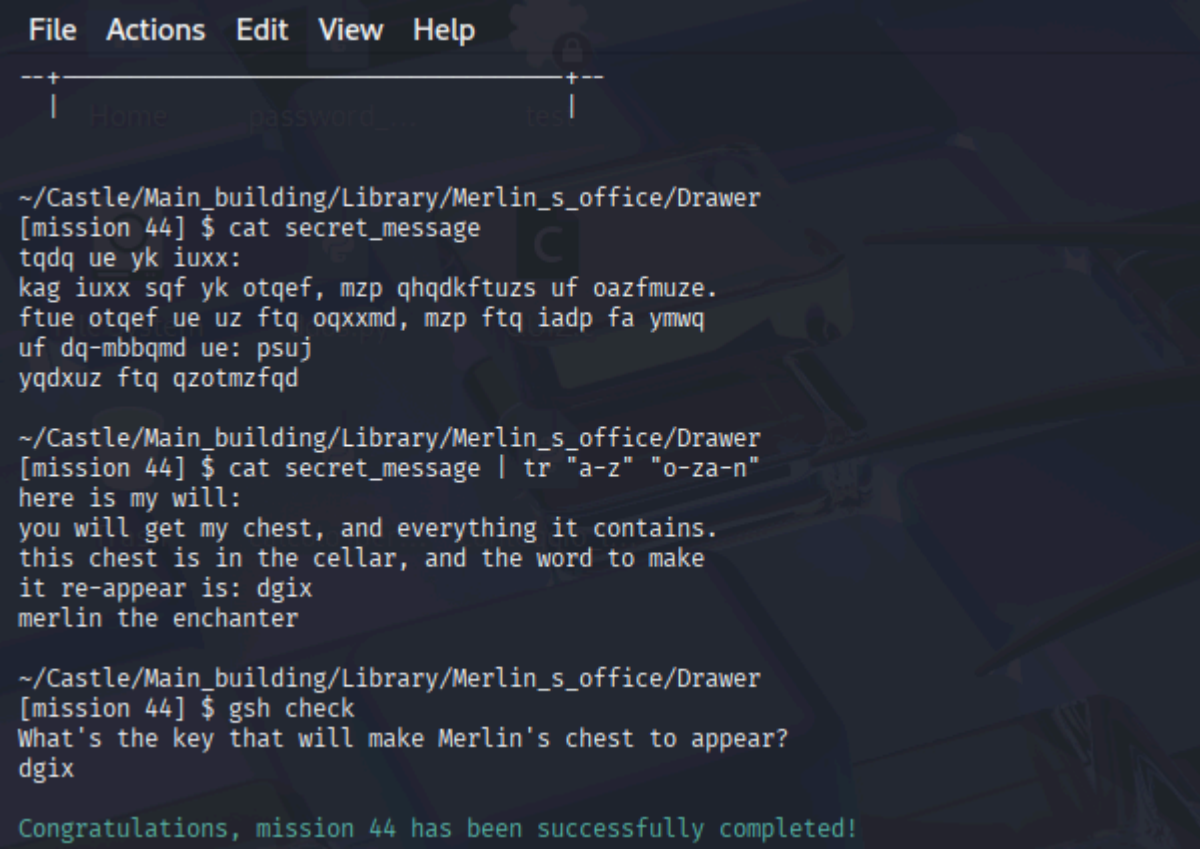


```
File Actions Edit View Help
--+-----+--
|                                     |
(0)
~/Stall
[mission 43] $ ls | grep -v 'boring_object' -v | xargs grep -v "PAID" | wc
61      532    2966
(1)
~/Stall
[mission 43] $ gsh check
How many unpaid items are there? 61
Congratulations, mission 43 has been successfully completed!
```

#### Missione n. 44:

Si richiede di decodificare un messaggio contenuto nell' ufficio di Merlino. Il presente "secret\_message" risulta cifrato con l'utilizzo di "Caesar shift cipher", ovvero una tipologia di cifrario che sostituisce una lettera con un'altra che si trova ad un numero di posizioni rispetto a quella da cifrare.

Per completare questa missione è necessario utilizzare il comando **tr** che esegue la sostituzione delle lettere con altre lettere che si trovano ad una determinata posizione rispetto alle prime. Nel caso specifico, dopo numerose prove, il risultato mostra che il comando **tr "a-z" "o-za-n"** risulta essere corretto in quanto mostra il messaggio segreto che rivela la parola chiave: **dgix**



```
File  Actions  Edit  View  Help
--+-----+--
| Home | password |
|-----|-----|

~/Castle/Main_building/Library/Merlin_s_office/Drawer
[mission 44] $ cat secret_message
tqdq ue yk iuxx:
kag iuxx sqf yk otqef, mzp qhqdktuzs uf oazfmuze.
ftue otqef ue uz ftq oqxxmd, mzp ftq iadp fa ymwq
uf dq-mbbqmd ue: psuj
yqdxuz ftq qzotmzfqd

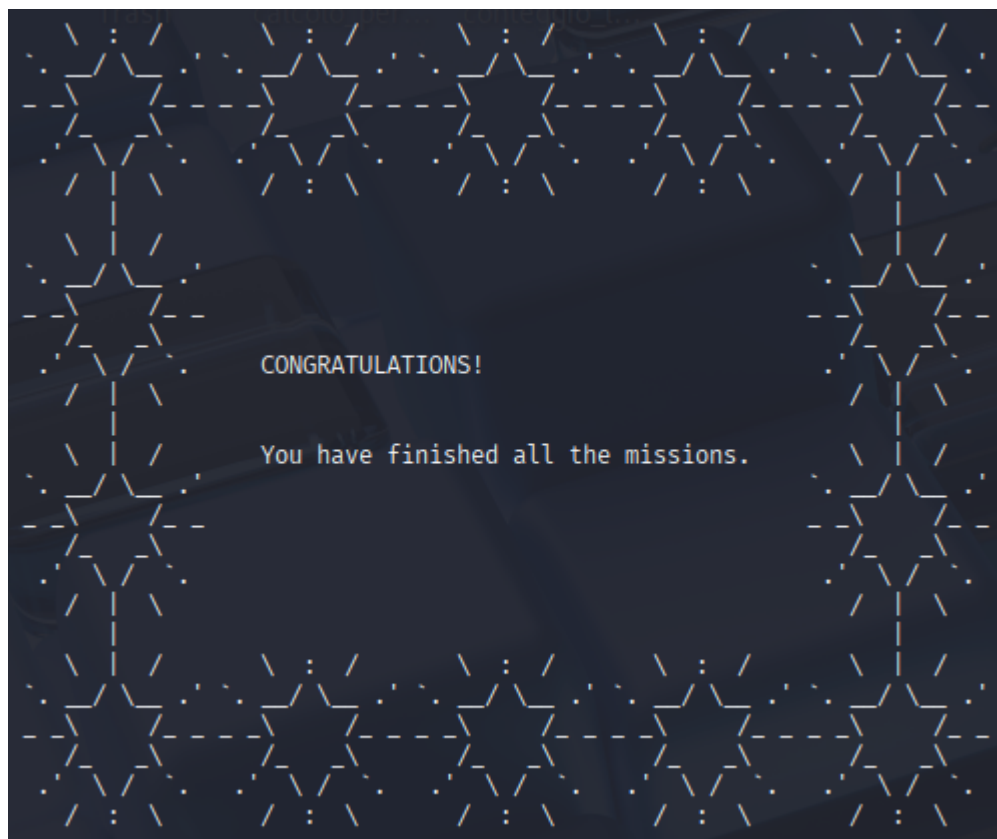
~/Castle/Main_building/Library/Merlin_s_office/Drawer
[mission 44] $ cat secret_message | tr "a-z" "o-za-n"
here is my will:
you will get my chest, and everything it contains.
this chest is in the cellar, and the word to make
it re-appear is: dgix
merlin the enchanter

~/Castle/Main_building/Library/Merlin_s_office/Drawer
[mission 44] $ gsh check
What's the key that will make Merlin's chest to appear?
dgix

Congratulations, mission 44 has been successfully completed!
```



La schermata seguente mostra il completamento di tutte le missioni.



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**Roma, 18/11/2025**